



Leveraging evolutionary algorithms with a dynamic weighted search space approach for fraud detection in healthcare insurance claims

Mohammad Tubishat ^{a,*,}, Dina Tbaishat ^{b,}, Ala' M. Al-Zoubi ^{c,}, Abed-Elalim Hraiz ^{d,},
Maria Habib ^e

^a College of Technological Innovation, Zayed University, Abu Dhabi, United Arab Emirates

^b College of Technological Innovation, Zayed University, Dubai, United Arab Emirates

^c Department of Data Science and Artificial Intelligence, Faculty of Science and Information Technology, Al-Zaytoonah University of Jordan, Amman, Jordan

^d King Abdullah II School of Information Technology, The University of Jordan, Amman, Jordan

^e Doctoral School of Sciences, Technologies, and Engineering, the University of Granada, Granada, Spain

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ABSTRACT

The healthcare industry has been suffering from fraud in many facets for decades, resulting in millions of dollars lost to fictitious claims at the expense of other patients who cannot afford appropriate care. As such, accurately identifying fraudulent claims is one of the most important factors in a well-functioning healthcare system. However, over time, fraud has become harder to detect because of increasingly complex and sophisticated fraud scheme development, data unpreparedness, as well as data privacy concerns. Moreover, traditional methods are proving increasingly inadequate in addressing this issue. To solve this issue a novel evolutionary dynamic weighted search space approach (DW-WOA-SVM) is presented in the current study. The approach has different levels that work simultaneously, where the optimization algorithm is responsible for tuning the Support Vector Machine (SVM) parameters, applying the weighting procedure for the features, and using a dynamic search space to adjust the range values. Tuning the parameters benefits the performance of SVM, and the weighting technique makes it updated with importance and lets the algorithm focus on data structure in addition to optimization objectives. The dynamic search space enhances the search range during the process. Furthermore, large language models have been applied to generate the dataset to improve the quality of the data and address the lack of good dimensionality, helping to enhance the richness of the data. The experiments highlighted the superior performance of this proposed approach than other algorithms.

1. Introduction

Healthcare fraud detection has become increasingly important due to rising costs and complexities associated with fraudulent claims. As healthcare systems expand and data volumes increase, accurately identifying fraudulent activities is crucial for maintaining financial integrity and ensuring appropriate resource allocation [1]. Fraudulent claims lead to significant financial losses for insurance providers and disrupt the delivery of essential services to genuine patients, impacting public health. For example, in a very recent event, the Montana Hospital agrees to pay 10.8M dollars to settle false claims in an oncology case [2]. Furthermore, the U.S. Justice Department announced in June a major crackdown, charging nearly 200 individuals involved in healthcare fraud schemes, with false claims amounting to more than 2.7 billion dollars [3]. Additionally, another incident reported by the CBS

news network stated that nationwide Medicare scams have resulted in over 100 million dollars in fake claims [4].

These cases demonstrate the importance of detecting such medical fraud. Detection of anomalies has been applied in various domains such as finance [5,6], cybersecurity [7–9], healthcare [10–13] and other areas [14–16]. Therefore, researchers, especially in healthcare, have been continuously developing new detection methods each year, using either traditional approaches or advanced techniques. However, traditional fraud detection methods struggle with the vast and complex nature of healthcare data, where anomalies and fraudulent patterns are subtle and diverse. Additionally, the imbalance between fraudulent and non-fraudulent claims can skew the performance of conventional detection models. Therefore, advanced analytical techniques are needed to

* Corresponding author.

E-mail addresses: Mohammad.Tubishat@zu.ac.ae (M. Tubishat), Dina.Tbaishat@zu.ac.ae (D. Tbaishat), a.alzoubi@zu.edu.jo (A.M. Al-Zoubi), abd8200493@ju.edu.jo (A.-E. Hraiz), mhabib@correo.ugr.es (M. Habib).

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handle high-dimensional data, identify complex patterns, and adapt to evolving fraud schemes.

Advancements in machine learning and optimization techniques offer promising solutions to these challenges. Bio-inspired optimization algorithms, such as Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) [17], Passing Vehicle Search (PVS), [18], Heat Transfer Search (HTS) [19], Multi-Objective Multi-Versé Optimization (MOMVO) [20], Cheetah Optimizer (CO) [21] have shown significant potential in enhancing the performance of machine learning models. One of the most popular classifiers in the literature is Support Vector Machines (SVMs). The algorithm uses a hyperplane to separate the classes, which allows it to excel over most algorithms, especially in detection problems. However, SVM performance relies on its parameters; therefore, using optimization algorithms can help address this issue.

Integrating these bio-inspired approaches with SVMs can improve feature selection and parameter tuning processes [22,23], leading to more accurate and robust fraud detection systems [24]. Another study applied SVM with optimization or evolutionary algorithms for health claim insurance prediction [25]. The authors used SVM with PSO to optimize parameters and improve health claim insurance identification performance, where PSO was implemented to optimize the SVM parameters. Applying feature selection and parameter tuning using optimization algorithms can improve the effectiveness of SVMs to some extent. Further, using a dynamic search space approach helps make the optimization process more efficient in dealing with complex problems. In other words, exploring a broader range of potential solutions can further improve the model's ability to achieve higher accuracy, especially with high-dimensional datasets.

Nevertheless, the lack of datasets in any field could affect the performance of any machine learning algorithm, especially in fraud detection for medical claims, particularly when data with high-dimensional is required. Therefore, the use of Large Language Models (LLMs) to generate data can help address this problem. LLMs are advanced artificial intelligence models that have already been trained on a large amount of text data to perform various tasks, especially in data generation techniques [26]. Tasks such as synthetic data generation can also be done by LLMs to improve the data quality [27]. For example, the authors of [28] utilized the data synthesis to rephrase web-text, while the work in [29] generate training data for classification matters. So, generating data with LLMs is not a new concept; however, it is increasingly being explored by researchers for its potential to enhance machine learning models, particularly in domains where datasets are lacking [30].

In this study, we utilized LLMs to generate data by following the dimensionality (features) of real datasets in medical fraud claims. Following such a standard helps the generated data to be more closely resemble real data with high-dimensional. Thus, increasing the number of features assists the models in identifying more patterns and ultimately improves their performance.

Therefore, to address the contribution of this work, we first implement a dynamic search space approach combined with a weighted evolutionary SVM for medical fraud claims detection. Metaheuristic algorithms are applied for parameter tuning, feature weighting, and dynamic search space adjustment. The dynamic search space helps to improve and adjust the search range of the parameters during the tuning process, while the weighting techniques allow the algorithm to adapt to the importance of each feature, focusing more on the structure of the data and the optimization objectives. Applying both techniques can achieve excellent performance in detecting medical fraud claims. Secondly, LLMs were applied to generate datasets with high-dimensional features and compare them with two real datasets.

Moreover, the experimental phase of this work consists of three phases, each with a varying number of parts. The first phase begins by evaluating the datasets using standard classification models, both with and without filter feature selection. The second phase, which involves

metaheuristic algorithms, consists of three parts: metaheuristic algorithms without feature selection (WFS), metaheuristic algorithms with feature selection (FS), and evaluating the proposed dynamic Weighted Search Space Approach using metaheuristic algorithms. In the third phase, the proposed approach is compared with the weighted approach using well-known benchmarks. Finally, a discussion of the improvements in feature performance is also presented.

In other words, the challenges of healthcare fraud detection, including high-dimensional datasets and imbalanced data distributions, necessitate advanced optimization techniques. Traditional metaheuristic algorithms often operate within fixed search spaces, which limit their ability to explore diverse regions of the solution space effectively. This static nature increases the likelihood of getting trapped in local optima, especially in scenarios where the solution space is complex and highly non-linear.

To address this limitation, our study introduces a dynamic search space approach. This innovative method dynamically adjusts the search boundaries based on the quality of solutions found during the optimization process. This dynamic adaptability ensures that the algorithm can:

- Enhance Exploration in Initial Stages: By widening the search boundaries at the beginning, the algorithm explores a broader range of potential solutions, reducing the risk of missing global optima.
- Refine Exploitation Near Promising Solutions: As the optimization progresses, the search space is narrowed to focus on regions with higher-quality solutions, improving convergence and ensuring robust performance.
- Handle High-Dimensionality and Imbalanced Data: By integrating this dynamic mechanism with a feature weighting system, the approach prioritizes the most influential features, effectively addressing the complexities of fraud detection datasets.

Preliminary experiments in our study demonstrated that the dynamic search space significantly improves convergence speed and solution quality, outperforming static search space methods. This evidence underscores its necessity in optimizing models for complex tasks such as healthcare fraud detection.

The remaining parts of the paper is structured as follows: A brief summary of the previous studies is given in 2 section. Section 3 presents a background of the used methods. The description of dataset is declared in Section 4. The methodology is explained in Section 5. The Experiments and Result of this research are discussed in Section 6. The final conclusion of this study is stated in Section 7.

2. Related work

In recent years, machine learning-based methods have increasingly been employed to tackle fraud detection challenges, leveraging both supervised and unsupervised learning techniques to enhance accuracy and efficiency. Bio-inspired algorithms have also gained prominence in fraud detection by integrating optimization techniques modeled after natural processes. The following sub-sections present recent work related to fraudulent healthcare claim detection.

2.1. Machine learning-based detection methods

Generally, fraud detection methods are divided into supervised and unsupervised approaches. Supervised methods require labeled training data, while unsupervised methods aim to find anomalies or abnormal patterns. Authors in [31] presented a survey of fraudulent behaviors detection, identifying data sources and characteristics, discussing key data preprocessing steps, and summarizing various statistical methods for fraud detection. Such techniques include neural networks, decision trees, association rules, Bayesian networks, and genetic algorithms.

Access to affordable healthcare is a widespread issue in the United States. In [32], the authors presented a Medicare fraud detection system using neural networks and conducted six experiments to resolve the problem of imbalanced data. The study emphasized optimizing decision thresholds and achieved a receiver operating characteristic curve (AUC) of approximately 85%. Hancock et al. [33] investigated the performance of CatBoost in detecting Medicare fraud using claims data with numerous categorical features. It demonstrated that CatBoost improved performance, achieving a mean AUC of 0.7851 compared to XGBoost's 0.7615 and further enhanced performance to a mean AUC of 0.8902 when including an additional categorical feature (provider state). This indicated that CatBoost was a superior alternative to XGBoost for Medicare fraud detection, particularly with categorical data. Additionally, Zhang et al. [34] created a model to identify abnormal healthcare records using outlier detection and a multi-label prediction method for disease-prescription correlation scores. This model focuses on spotting inconsistencies in healthcare data through advanced predictive methods using deep convolution and recurrent models. Limiting the dataset to fewer than 1000 drugs and diseases improved precision but reduced anomaly detection to 71% of records. Privacy concerns led to insufficient training data, impacting the model's robustness. Matloob et al. [35] introduced a novel methodology for detecting insurance claim fraud in healthcare using sequence mining and prediction techniques. Unlike previous methods that focused on monetary or medication-disease analysis, this approach generated and analyzed sequences of services provided by different specialties. It involved a "Sequence rule engine" to identify frequent and rare sequences and a "Prediction-based engine" to further analyze anomalies. Validated with data from a local hospital, the methodology achieved an average accuracy of 85%, proving effective in identifying fraudulent claims and enhancing patient management. In [36], the authors analyzed various machine learning techniques for identifying Medicare fraud, focusing on the isolated forest algorithm, which improved performance by detecting outliers. The study demonstrated that the isolated forest algorithm achieved approximately 98.76% accuracy, showing promising results compared to existing fraud detection methods. Moreover, Johnson et al. [37] explored four techniques for encoding medical provider types into dense, semantic embeddings to enhance healthcare fraud detection. The methods included two pre-trained word embedding techniques and two novel methods that utilize procedure and drug occurrence data, all of which significantly improve fraud classification accuracy when tested on real-world datasets. Kapadiya et al. [38] surveyed AI and blockchain technologies for enhancing fraud detection in health insurance, presenting a taxonomy of security issues. Also, it proposed a four-layer intelligent fraud detection system that was utilized for the case of how health insurance firms use wearable device data to promote healthy behaviors. The increasing rate of healthcare fraud, exacerbated by the coronavirus pandemic, highlights the need for effective detection methods. Kalra et al. [39] introduced a fuzzy expert system designed to identify insurance fraud accurately and quickly, addressing the complex and costly nature of fraud in the insurance sector. In [40], a novel process-based fraud detection methodology using sequence mining concepts was proposed. In this method, Matloob et al. generated frequent sequences of services within each specialty and compared them with actual patient data to identify anomalies. Validated with five years of transactional data from a local hospital, the methodology effectively identified fraudulent activities and aimed to improve transparency and optimize costs and benefits in healthcare systems.

Also, authors in [41] developed Medicare fraud detection models employing two approaches: graph neural network (GNN) models and traditional machine learning models using graph centrality measures. The machine learning model with graph centrality features outperformed the best GNN model by obtaining an f1-score of 65%. The best GNN model required 250–300 times more training time than the best machine learning model, suggesting that leveraging graph

centrality measures can lead to effective Medicare fraud detection. Nonetheless, [1] proposed a machine learning approach utilizing Random Forest (RF), Logistic Regression (LR), and Artificial Neural Networks (ANN) for fraud detection in healthcare insurance claims. The authors achieved the best results with the RF classifier achieving an f1-score of 99.03%. In [42], the study presented six new labeled Medicare datasets derived from publicly available CMS data, focusing on enhancing healthcare fraud classification. The new "Summary by Provider" features and "Aggregated-Enriched" datasets significantly improve fraud detection. For the "Aggregated-Enriched", the XGBoost algorithm attained an average g-mean of 90.81%. Lu et al. [43] designed the MHAMFD, a health insurance fraud detection model that used a multilevel hierarchical attention mechanism with GNN to examine behavioral patterns across multiple patient visits. Experimental findings showed that the proposed model performed better than conventional methods in identifying fraud by having an f1-score of 0.8439. This highlighted the significance of considering patient visit behavior in fraud detection. Fraud in the United States healthcare system causes significant financial losses and impacts beneficiaries' well-being. Shi et al. [44] introduced a fraud detection method using Medicare claims data, focusing on temporal information from previous years. The approach used functional principal component analysis (FPCA) for feature engineering and cost-sensitive learning to address the class imbalance and optimize the trade-off between fixed and fraud costs, resulting in improved prediction performance and substantial cost savings. Furthermore, Mao et al. [45] focused on creating a medical insurance fraud detection model using machine learning to address issues with imbalanced fraudulent and normal samples. It employed traditional over-sampling, under-sampling, and new samples generated by Generative Adversarial Networks (GANs) to tackle sample imbalance, which improved binary classification effectiveness. The fraud monitoring model was built using both LR and XGBoost models. The results showed that over-sampling and GAN methods better preserved sample characteristics compared to under-sampling, and the XGBoost model outperformed the LR model in classification and detection by obtaining an f1-score of 95.4%.

2.2. Bio-inspired based detection methods

One of the earliest approaches was in [46], in which the K-nearest neighbor (KNN) algorithm was combined with a genetic algorithm to address a medical fraud detection problem. The genetic algorithm was used to find optimal feature weights for classifying General Practitioners' (GP) practice profiles. These weights were then applied in the KNN algorithm to identify nearest neighbor profiles, with classifications determined using the majority rule and Bayesian rule. The results demonstrated that this approach achieved effective generalization in classifying GP practice profiles, paving the way for its application in medical fraud detection at the Health Insurance Commission (HIC). In [24], Sowah et al. developed a decision support system for automated and accurate claim processing using data from the National Health Insurance Scheme in Ghana. Genetic Support Vector Machines (GSVMs), a hybrid tool combining data mining and machine learning, were employed to detect fraudulent claims. The GSVM demonstrated superior detection and classification performance compared to traditional methods, reducing computational time and improving classification accuracy with SVM kernel classifiers: linear (80.67%), polynomial (81.22%), and radial basis function (RBF) kernel (87.91%). Also, [47] Majhi et al. proposed a hybrid fuzzy clustering technique using the Salp Swarm Algorithm (SSA) to optimize cluster centroids and eliminate outliers through undersampling. This method was compared with other clustering algorithms, demonstrating its superiority. A novel automobile insurance fraud detection system was developed, where the proposed fuzzy clustering method balanced the dataset by removing outliers from majority class samples. The balanced dataset was then classified using RF, LR, and XGBoost classifiers. Among these, the

combination of the fuzzy clustering method and XGBoost classifier achieved the best performance in terms of sensitivity, accuracy, and specificity.

Later on, as the insurance industry developed, insurance fraud increased rapidly, significantly hindering the industry's progress. In [48], the authors proposed an improved adaptive genetic algorithm (NAGA) combined with a BP neural network to enhance fraud detection. The NAGA-BP model aimed to optimize the initial weights of the BP neural network, overcoming issues like local minima, slow convergence, and sample dependence. Using historical automobile insurance claim data for simulation and prediction, the results demonstrated that the NAGA-BP model outperformed traditional genetic algorithms in terms of convergence speed and prediction accuracy. Insurance fraud was identified as one of the most expensive financial crimes, involving deliberate illegal payments of benefits. Parnian et al. [49] employed a combination of feature subset selection using a squirrel optimization algorithm and nearest neighbor classification to detect fraud in health insurance. Using educational data from Ardabil's Social Security Insurance Organization, the method effectively eliminated irrelevant features, improving the accuracy of fraud detection compared to other classification algorithms. Detecting healthcare fraudulent transactions is challenging due to the complex interactions among doctors, patients, and services. Moreover, Majhi et al. [50] proposed a fuzzy clustering method that combined the strengths of the modified whale optimization algorithm (MWOA) and fuzzy c-means (FCM) to improve clustering performance. While FCM is widely used, it often gets stuck in local optima, but MWOA, a stochastic global optimization algorithm, was modified to address this issue. The effectiveness of the hybrid clustering method was assessed using various metrics and applied to an automobile insurance fraud detection system (AIFDS). The AIFDS utilized the proposed clustering method to trim outliers from the dataset before applying advanced classifiers like CATBoost, XGBoost, RF, LightGBM, and Decision Tree. The results showed that the AIFDS with MWOA-based fuzzy clustering and CATBoost outperformed the other methods in terms of sensitivity, specificity, and accuracy.

Additionally, in response to the risks of fraudulent online financial transactions, Lingeswari et al. [51] developed DEGA (Deep Emended Genetic Algorithm) to enhance detection performance. DEGA improved accuracy through new crossover and selection methods, dynamic mutation functions, and a competitive approach. Additionally, EL-UXGB (Efficient Loss-Updated eXtreme Gradient Boosting) was introduced to tackle imbalanced labels with dual sigmoid loss functions. Both methods demonstrated effective performance in detecting fraudulent transactions. Table 1 summarizes the works discussed in the related work section.

3. Preliminaries

In this section, we describe the background of the methods and techniques used in this study. These methods include Large Language Models, Support Vector Machines, and the Whale Optimization Algorithm.

3.1. Large Language Models (LLMs)

Large Language Models (LLMs) are considered a subset of artificial intelligence (AI) systems that are designed to understand, generate, and process human language as accurately as possible. These models, such as GPT-4o, have been trained on a huge number of datasets that contain different text sources to learn semantics, syntax, and linguistic patterns. Such models usually operate by using deep learning architectures, primarily transformers, which have the ability to capture long-range dependencies and contextual information in text (See Fig. 1). Accordingly, LLMs are regarded as excellent for handling natural language processing (NLP) tasks, including sentiment analysis, machine translation, text summarization, and question-answering.

One key feature of LLMs is their ability to generalize knowledge across domains, making them versatile tools for applications such as chatbots, content generation, and automated code completion. Furthermore, their capacity to use the pre-training and fine-tuning process allows them to adapt to specific tasks or domains with relatively small amounts of task-specific data. In recent research, LLMs have also been used in more specialized fields such as healthcare, where they help in medical report analysis, diagnosis support, and even fraud detection in insurance claims.

Regardless of their significant capabilities, LLMs face a number of challenges, namely biases present in training data, high computational costs, and the need for large-scale infrastructure. Their "black-box" nature also raises concerns regarding interpretability and accountability in decision-making processes. However, continuous improvements help them reduce and eliminate these challenges, especially in ethical and practical deployment.

3.2. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for solving both classification and regression problems. Initially developed by Vladimir Vapnik, SVM functions as a nonlinear binary classifier [52,53]. SVM is particularly effective at transforming nonlinear data into a form that can be linearly separated. It operates by constructing a hyperplane that maximizes the margin between two different classes of data points. This hyperplane acts as a decision boundary that separates the classes, ensuring correct division between them. Moreover, SVM can be used for both linear and nonlinear relationships in data. In addition to classification tasks, SVM can also function on regression problems by creating hyperplanes that provide linear separation within a high-dimensional vector space as shown in Fig. 2.

Each data point consists of a pair (x_i, y_i) , where x_i is a feature vector represented as $(x_{i1}, x_{i2}, \dots, x_{ip})$, with p denoting the number of features and $i = 1, \dots, n$ representing the individual instances in the dataset. Further, n refers to the number of training instances, while y_i represents the class label assigned to each instance [54]. To effectively separate the data points across classes, the feature space must be partitioned. A hyperplane is considered optimal and achieves the best results when it maximizes the margin between the hyperplane and the nearest training point. Support vectors (SVs) are the observations located at a distance equal to or greater than one margin from the hyperplane. These vectors are situated in the feature space, and their positions determine the placement of the hyperplane. Thus, any shift in the position of these observations leads to a change in the hyperplane [54].

For optimal classification, hyperplanes ideally should separate data points into distinct classes in a linear way. However, in real-world scenarios, achieving perfect separation is often challenging, leading to potential misclassifications with linear models. To solve this, the original feature space can be mapped into a higher-dimensional space (ϕ), which eventually allows for easy discrimination of data points. This process relies on kernel functions to expand the space nonlinearly, enabling better class separation. Several types of kernel functions can be used, including:

- Linear kernels:

$$k(x_i, x'_i) = \sum_{j=1}^p x_{ij} x'_{ij} \quad (1)$$

- Polynomial kernels:

$$k(x_i, x'_i) = (1 + \sum_{j=1}^p x_{ij} x'_{ij})^d \quad (2)$$

- RBF kernels:

$$k(x_i, x'_i) = \exp(-\gamma \sum_{j=1}^p (x_{ij} - x'_{ij})^2) \quad (3)$$

Table 1
Summary of related work in medical fraud detection.

Approach	Methodology	Results
Li et al. (2008)	Survey of fraudulent behavior detection with methods such as neural networks, decision trees, Bayesian networks, and genetic algorithms.	Discusses key preprocessing and statistical methods for fraud detection.
Johnson et al. (2019)	Medicare fraud detection using neural networks and imbalanced data handling.	Achieved an AUC of 85%.
Hancock et al. (2020)	Medicare fraud detection using CatBoost on claims with categorical data.	Improved AUC from 0.7851 (XGBoost) to 0.8902 with additional categorical features.
Zhang et al. (2020)	Outlier detection for healthcare fraud using multi-label prediction.	Improved precision, but anomaly detection dropped to 71% due to dataset limitations.
Matloob et al. (2020)	Sequence mining and prediction techniques for insurance claim fraud detection.	Achieved 85% accuracy, focusing on service sequences.
Bhaskar et al. (2021)	Medicare fraud detection using the isolated forest algorithm.	Achieved 98.76% accuracy, showing promise in identifying outliers.
Johnson et al. (2021)	Fraud detection enhanced by medical provider-type embeddings.	Significant improvement in classification accuracy using real-world datasets.
Kapadiya et al. (2022)	AI and blockchain for enhancing fraud detection in health insurance using wearable data.	Discussed security issues and proposed a four-layer fraud detection system.
Kalra et al. (2022)	Fuzzy expert system for insurance fraud detection.	Enhanced accuracy and speed in identifying insurance fraud.
Yoo et al. (2023)	Medicare fraud detection using graph neural networks (GNN) and traditional models with graph centrality measures.	Machine learning model with graph centrality outperformed GNN with an F1-score of 65%.
Nabrawi et al. (2023)	Medicare fraud detection using Random Forest, Logistic Regression, and Artificial Neural Networks (ANN).	Random Forest achieved an F1-score of 99.03%.
Johnson et al. (2023)	Six new labeled Medicare datasets and fraud classification with XGBoost.	XGBoost achieved a g-mean of 90.81%.
Lu et al. (2023)	Fraud detection using a multilevel hierarchical attention mechanism with GNN.	Achieved an F1-score of 0.8439, highlighting patient visit behavior significance.
Shi et al. (2023)	Medicare fraud detection with functional principal component analysis (FPCA) and cost-sensitive learning.	Improved prediction performance and cost savings.
Mao et al. (2024)	Fraud detection using Generative Adversarial Networks (GAN) for imbalanced data.	XGBoost with GAN achieved an F1-score of 95.4%.
He et al. (1999)	K-nearest neighbor (KNN) with a genetic algorithm for medical fraud detection.	Demonstrated effective generalization in classifying GP profiles.
Sowah et al. (2019)	Genetic Support Vector Machines (GSVMs) for detecting healthcare fraud in Ghana's National Health Insurance Scheme.	Achieved classification accuracy up to 87.91% using RBF kernel.
Majhi et al. (2019)	Fuzzy clustering with Salp Swarm Algorithm (SSA) for fraud detection in automobile insurance.	SSA and XGBoost achieved the best performance in sensitivity, accuracy, and specificity.
Yan et al. (2020)	Adaptive genetic algorithm (NAGA) combined with BP neural networks for fraud detection in automobile insurance.	Improved convergence speed and prediction accuracy.
Parnian et al. (2021)	Squirrel optimization and nearest neighbor classification for healthcare fraud detection.	Improved fraud detection accuracy by eliminating irrelevant features.
Majhi et al. (2021)	Fuzzy clustering with modified whale optimization and fuzzy c-means for fraud detection in automobile insurance.	Outperformed other methods in sensitivity, specificity, and accuracy with CATBoost.
Lingeswari et al. (2024)	DEGA (Deep Emended Genetic Algorithm) and EL-UXGB (Efficient Loss-Updated eXtreme Gradient Boosting) for fraud detection in online financial transactions.	Enhanced accuracy in detecting fraudulent transactions, particularly with imbalanced data labels.

In this context, $k(\cdot)$ represents the kernel function, and x_i and x'_i are the observation vectors used in the inner product calculation. The inner product is denoted by $\varphi(x_i) \cdot \varphi(x'_i)$. So, to illustrate, the SVM performance largely depends on its parameters (γ and C) and the kernel function. In this work, we use the Radial Basis Function (RBF) kernel to manage nonlinear variables and decision boundaries. The RBF kernel is computed using Eq. (3). As for the SVM parameters, the Gamma coefficient (γ) controls the influence of individual data points, while the cost parameter (C) acts as a penalty term to improve classification accuracy. It is worth mentioning that improper setting of γ and C can result in poor generalization, leading to either underfitting or overfitting.

3.3. Whale Optimization Algorithm (WOA)

The Whale Optimization Algorithm (WOA) is a search and optimization technique introduced by [55] and inspired by the hunting behavior of humpback whales that follow the bubble-net feeding strategy. By employing this strategy, the whales dive approximately 12 meters deep and create spiral-shaped bubbles around their prey. Then, the whales swim upward to capture the fish as shown in Fig. 3.

Therefore, the WOA follows this strategy during the exploitation process, where the algorithm targets the optimal solutions similar to how whales hunt. As for exploration, the WOA mimics the whales' random search for food by adjusting solutions based on the relative positions of other whales. Furthermore, the algorithm does not always

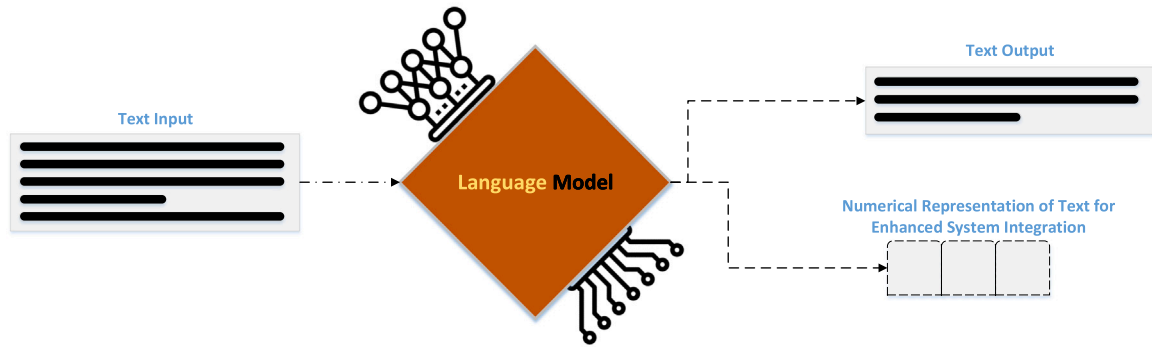


Fig. 1. Large language model.

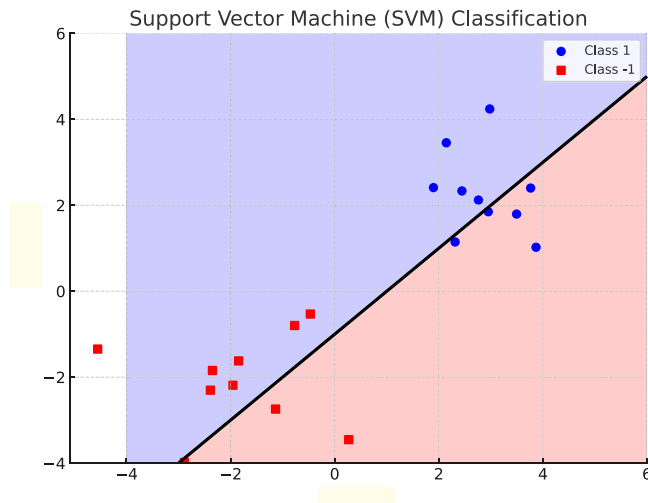


Fig. 2. Support vector machine.



Fig. 3. Bubble-net whale hunting behavior.

try to select the best solution; instead, it occasionally picks random alternatives, allowing for a more diverse search. The WOA stands out from other optimization algorithms given its simplicity — only two parameters need adjustment. This results in an easier way to balance exploration and exploitation.

The mathematical representation of the WOA begins its operation by randomly initializing a set of potential solutions, as described in Algorithm 1. Further, the algorithm calculates the fitness value of each solution and determine the best one while store its position in S^* . Then the positions of the search agents are adjusted based on that. The criteria that the whales use to update their positions in either a circular or spiral motion. So, in the algorithm, If the probability p – a random number between 0 and 1 – is greater than or equal to 0.5, the position of the current agent is updated using the following equation:

$$\vec{S}(t+1) = \vec{D}' \cdot e^{bl} \cdot \cos(2\pi l) + \vec{S}^*(t) \quad (4)$$

This equation represents the spiral-shaped movement, where $\vec{S}$ is the position vector, t is the current iteration, $\vec{D}'$ is the distance between the current and best search agents, b is a constant that defines the shape of the spiral, and l is a random number between -1 and 1 .

Moreover, if p is less than 0.5, the algorithm updates the search agent's position based on a circular hunting pattern. However, before updating the position of the search agent, the algorithm checks whether the coefficient vector $\vec{A}$ falls within the range $(-1, 1)$. If it does, the algorithm adjust the position of the current agent to the best search agent bead on Eq. (5).

$$\vec{S}(t+1) = \vec{S}^*(t) - \vec{A} \cdot \vec{D} \quad (5)$$

Otherwise, if $\vec{A}$ falls outside the range $(-1, 1)$, the position of the current search agent is updated in a circular motion based on a randomly chosen search agent, as shown in Eq. (6). In the equation, $\vec{S}_{rand}$ represents the position vector of the randomly selected agent, and $\vec{D}$ denotes the distance between the current and the random search agents.

$$\vec{S}(t+1) = \vec{S}_{rand} - \vec{A} \cdot \vec{D} \quad (6)$$

After completing these steps for each search agent in an iteration, the fitness value of every search agent is calculated. If a search agent finds a better solution than the one currently stored in S , S is updated with the new best solution. Eventually, the process stops once the algorithm reaches the maximum number of iterations.

4. Dataset description, characteristics, and preparation

This section provides an overview of the datasets used in this study, highlighting their key characteristics and detailing the preparation steps undertaken before analysis.

We should note that our rule-generating standards on the medical fraud claim-based real data are a model to be used to generate proposed

Algorithm 1 Whale Optimization Algorithm (WOA)

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1: Input: Population size  $N$ ; Max iterations  $T_{\max}$ ; search space boundaries; fitness function  $f(x)$ 
2: Initialize whale population  $\{S_i\}_{i=1}^N$  within the search space
3: Evaluate fitness  $f(S_i)$  for all whales
4: Set  $S^* = \arg \min f(S_i)$ 
5: for  $t = 1$  to  $T_{\max}$  do
6:   for each whale  $S_i$  do
7:     Update parameters  $a$ ,  $A$ ,  $C$ ,  $l$ , and  $p$ 
8:     if  $p < 0.5$  then
9:       if  $|A| < 1$  then
10:        Update position  $S_i$  using Eq. (5)
11:       else
12:        Select a random whale  $S_{\text{rand}}$ 
13:        Update position  $S_i$  using Eq. (6)
14:       end if
15:     else
16:       Update position  $S_i$  using Eq. (4)
17:     end if
18:   end for
19: Ensure all whales remain within search boundaries
20: Evaluate fitness  $f(S_i)$  for all whales; update  $S^*$  if a better solution is found
21: end for
22: Return best solution  $S^*$ 

```

datasets, which are utilized through an LLM. These rules help to ensure that the output data looks more natural and similar in content and pattern to real-life fraud (for instance claim amounts, diagnosis codes, service dates etc.). In turn, the dataset is more realistic and can be better suited to training models for fraud detection.

The synthetic dataset was generated using an LLM in the following steps:

- **Feature Alignment Across Real Datasets:**

- The two real datasets were analyzed to identify common features and complementary attributes that represent key aspects of healthcare claims (e.g., demographic details, policy information, claim details, and healthcare provider metrics).
- A unified schema was developed to incorporate these features, ensuring a comprehensive representation of the data. This schema included 63 features, covering the essential aspects of both datasets and extending their dimensionality to capture more nuanced patterns.

- **Guiding the LLM for Data Generation:**

- Prompt engineering techniques were used to provide the LLM with detailed descriptions of the unified schema. Prompts included the types of features (categorical, numerical, boolean, etc.), their possible value ranges, and real-world examples extracted from the original datasets.
- The LLM generated data entries following these prompts, ensuring that the synthetic data adhered to the specified schema and mirrored real-world distributions.

- **Adding Realism and Diversity:**

- To enhance the realism of the synthetic dataset, the LLM was prompted to include contextual relationships between features. For instance, a high claim amount might correspond to a severe medical condition or a particular type of policy coverage.

- The generated data was further diversified by introducing variations in temporal and categorical attributes, creating a dataset that reflects real-world heterogeneity.

4.1. Dataset description

This study uses a dataset and includes an array of features broad enough to capture several important attributes such as insurance claims history, customer demographics, policy details, and the healthcare provider's operations. It could prove to be a useful repository indeed, for detecting fraud, risk analysis, and well health care services.

The 63-feature data set describes multiple aspects of the complete lifecycle of an insurance claim, from its creation through to final settlement. These features go across different domains:

- **Insurance Data** (Claim ID, Claim Date, Claim Amount, Claim Type, Approval Status, Settlement Amount) These components display details on each claim including the status of that claim within the processing flow.
- **Customer Demographics:** This category includes variables such as Customer Age, Gender, Marital Status, and Income etc., which are detailed information about the demographics of a customer submitting claims.
- **Policy Information** — Provides details about the policies associated with the claims, i.e., Policy Number, Policy Start Date, Policy End Date, and Coverage Type.
- **Healthcare Provider Metrics**—This is to capture the key operational aspects of the healthcare providers that are involved in the claims: (a) Number of Providers the doctor works for. (b) The average number of Patients seen per day. (c) Number of Consultations.
- **Patient Management** — The Total Number of Patients Per Month and the Number of Revisits Within 24 Hours with the Same Diagnosis are a couple of metrics that demonstrate how patients respond to medical care and utilize the healthcare system.

4.2. Dataset characteristics

The database contains various types of features providing a comprehensive overview of benchmarking results, customer demographics, information about insurance claims, and biometric measures bundled with our data. The different classifications of features would be;

- **Categorical Features:** The dataset contains the following categorical variables such as Claim Type, Customer Gender, Marital Status, Coverage Type, Claim Approval Status, and Fraudulent Status. Classification tasks in the study require these functionalities.
 - **Numerical Features:** There are plenty of numerical variables in the dataset such as Claim Amount, Customer Age, Income, and various healthcare metrics. In particular, these features are important for quantitative and statistical modeling.
 - **Temporal Features:** Variables oriented around time including Claim Date, Policy Start Date, and Settlement Date for interpreting the temporal dimension of claims and policy management.
 - **Boolean Features:** Such variables represent true or false presence of a specific condition (e.g., whether the claim is authored by someone known as trustworthy)
- The feature names and descriptions can be found in [Tables 2 and 3](#).

4.3. Data preparation

To prepare this dataset for analysis or machine learning, several preprocessing steps are necessary:

Table 2
Feature details part 1.

#	Feature	Description	Values
F1	Claim ID	Unique identifier for each insurance claim	Alphanumeric ID (e.g., CLM12345)
F2	Claim Date	Date when the claim was filed	Date (e.g., 2023-01-15)
F3	Claim Amount	The total monetary amount claimed	Numeric (e.g., \$1,500)
F4	Claim Type	Category of the claim (e.g., accident, health, property)	Text (e.g., Health, Auto, Property)
F5	Customer Age	Age of the customer filing the claim	Numeric (e.g., 45)
F6	Customer Gender	Gender of the customer	Text (Male/Female/Other)
F7	Customer Marital Status	Marital status of the customer	Text (Single, Married, Divorced)
F8	Customer Income	Annual income of the customer	Numeric (e.g., \$75,000)
F9	Policy Number	Unique identifier for the insurance policy	Alphanumeric ID (e.g., POL12345)
F10	Policy Start Date	Start date of the insurance policy	Date (e.g., 2022-01-01)
F11	Policy End Date	End date of the insurance policy	Date (e.g., 2023-01-01)
F12	Coverage Type	Type of coverage provided by the insurance policy	Text (e.g., Full, Partial)
F13	Number of Claims	Total number of claims filed by the customer	Numeric (e.g., 3)
F14	Number of Policies	Total number of policies held by the customer	Numeric (e.g., 2)
F15	Claim Approval Status	Status of the claim approval process	Text (Approved, Pending, Denied)
F16	Investigation Flag	Indicator of whether the claim is under investigation	Boolean (True/False)
F17	Settlement Amount	The amount settled for the claim	Numeric (e.g., \$1,200)
F18	Settlement Date	Date when the claim was settled	Date (e.g., 2023-02-01)
F19	Location of Incident	Location where the incident occurred	Text (City, State, Country)
F20	Claim Description	Description of the claim	Text
F21	Number of Providers This Doctor Works For	Total number of healthcare providers the doctor is affiliated with	Numeric (e.g., 5)
F22	Avg Number of Patients Per Day	Average number of patients treated by the doctor per day	Numeric (e.g., 20)
F23	Avg Number of Patients Per Week	Average number of patients treated by the doctor per week	Numeric (e.g., 100)
F24	Total Number of Patients Per Month	Total number of patients treated by the doctor per month	Numeric (e.g., 400)
F25	Standard Deviation of Number of Patients Per Month	Statistical measure of variability in the number of patients per month	Numeric (e.g., 15)
F26	Mode of Number of Patients Per Month	Most frequent number of patients treated by the doctor per month	Numeric (e.g., 35)
F27	Kurtosis of Number of Patients Per Month	Measure of the "tailedness" of the patient number distribution	Numeric (e.g., 3.5)
F28	Ratio of New Patients to Old Patients	Ratio of newly registered to returning patients	Numeric (e.g., 0.8)
F29	Total Number of Patients Per 1st Weekend in the Month	Total number of patients treated by the doctor on the first weekend	Numeric (e.g., 25)
F30	Total Number of Patients Per 2nd Weekend in the Month	Total number of patients treated by the doctor on the second weekend	Numeric (e.g., 30)
F31	Total Number of Patients Per 3rd Weekend in the Month	Total number of patients treated by the doctor on the third weekend	Numeric (e.g., 20)
F32	Total Number of Patients Per 4th Weekend in the Month	Total number of patients treated by the doctor on the fourth weekend	Numeric (e.g., 22)

4.3.1. Data cleaning

A number of steps were taken to preprocess the dataset before doing analysis and modeling so that it should be clean, and tidy by cleaning each column and defining some meaningful variable names. The first stage of the pipeline comprised data cleaning, with missing values being addressed through either imputation if feasible, or else removal of incomplete records to ensure the robustness of data. We have also treated numerically cascaded outliers (from Claim Amount, Settlement Amount, and some healthcare metrics) to avoid distortion of results. Temporal features had feature engineering applied to them to bring out components such as year, month, and day of the week (i.e., claim time and policy effective time) patterns.

Calculating ratios and differences between existing variables provided new features that quantify different aspects of both customer behavior (e.g. entityManager-to-dataNuker ratio, dataBitchperItm postPeak-prePeak) and healthcare provider performance. After which the data was further processed by getting numerical features to be within a similar scale. This is especially useful for algorithms sensitive to feature scaling, like support vector machines, so we could change this by normalizing or standardizing them. Categorical variables were transformed into machine learning-ready formats, with techniques like one-hot encoding or label encoding. The Fraudulent feature is also an issue of imbalance, to deal with it resampling, synthetic data generation, or specialized algorithms for imbalanced datasets were used. These comprehensive data preparation steps were essential for structuring the dataset in a way that supports robust analysis and reliable predictive modeling, ensuring the accuracy and reliability of the study's outcomes.

Our generated dataset consists of 18,000 instances with 63 features. The target class has two labels: Fraud (True) or Not Fraud (False). More details of the used datasets in this study can be found in Table 4.

5. The proposed approach

This section presents the implementation of the proposed approach by describing the details of each phase. Three phases need to be considered when designing our approach: the encoding scheme, the fitness function, and the system architecture. The dynamic search space

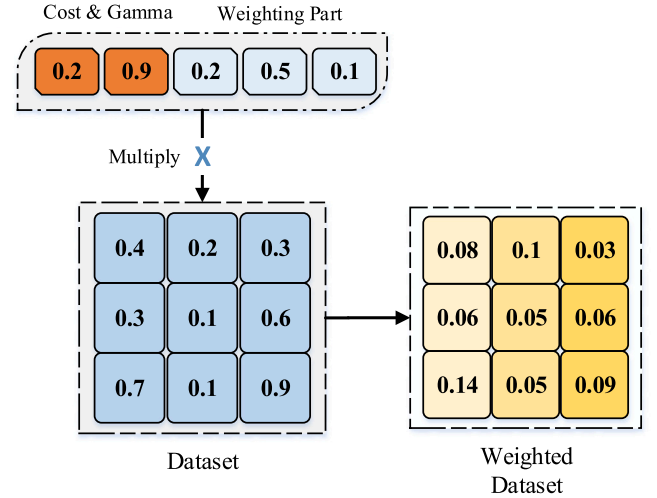


Fig. 4. Representation of the solution encoding in the proposed approach.

and the weighting of the features are both designed to improve the performance of the SVM for fraud detection. Furthermore, the evaluation process is explained to assess the performance of the approach.

In any model implementation, the solution representation must first be designed. In this work, the architecture of the encoding scheme consists of individuals (also known as particles) in the form of vectors of real numbers. In our case, two elements must be taken into account: the feature elements and the SVM parameter elements (Cost C and Gamma γ) shown in Fig. 4. In other words, the encoding scheme consists of the aforementioned two elements. It is worth mentioning that the vector elements are randomly generated numbers that will be decoded based on Eq. (7). To clarify further, WOA generates elements that encompass the two SVM parameters plus the D number of features for each dataset. The search space for these parameters and features will be dynamically adjusted throughout the optimization process.

$$B = \frac{A - \min_A}{\max_A - \min_A} (\max_B - \min_B) + \min_B \quad (7)$$

Table 3
Feature details part 2.

#	Feature	Description	Values
F33	Total Number of Patients Per Official Days Off in Seasonality	Number of patients treated during official holidays or seasonal breaks	Numeric (e.g., 15)
F34	Number of Revisit Within 24 Hours with the Same Diagnosis per Month	Number of patients who revisit within 24 h with the same diagnosis	Numeric (e.g., 5)
F35	Number of Revisit Within 48 Hours with the Same Diagnosis per Month	Number of patients who revisit within 48 h with the same diagnosis	Numeric (e.g., 3)
F36	Revisit for the Same Diagnosis But with Other Doctor	Number of patients revisiting for the same diagnosis but with another doctor	Numeric (e.g., 7)
F37	Percentage of This Doctor Using the Same Primary Diagnosis	Percentage of cases where the doctor uses the same primary diagnosis	Percentage (e.g., 75%)
F38	Percentage of This Doctor Using the Rare Primary Diagnosis	Percentage of cases where the doctor uses a rare primary diagnosis	Percentage (e.g., 5%)
F39	Percentage of This Doctor Using the Same Medication	Percentage of cases where the doctor prescribes the same medication	Percentage (e.g., 65%)
F40	Percentage of This Doctor Using the Rare Medication	Percentage of cases where the doctor prescribes a rare medication	Percentage (e.g., 10%)
F41	Percentage of This Doctor Using the Same Nphies Standard Description	Percentage of cases using the same standardized description in documentation	Percentage (e.g., 60%)
F42	Percentage of This Doctor Using the Rare Nphies Standard Description	Percentage of cases using rare standardized descriptions in documentation	Percentage (e.g., 3%)
F43	Number of Consultations	Total number of consultations carried out by the doctor	Numeric (e.g., 120)
F44	Number of Cases Dental	Total number of dental cases handled by the doctor	Numeric (e.g., 45)
F45	Total Number of Drugs	Total number of different drugs prescribed by the doctor	Numeric (e.g., 50)
F46	Total Number of Medical Devices	Total number of different medical devices prescribed or used	Numeric (e.g., 10)
F47	Total Number of Services	Total number of different medical services provided	Numeric (e.g., 75)
F48	Number of High-Cost Services	Total number of high-cost services provided	Numeric (e.g., 5)
F49	Number of High-Cost Drugs	Total number of high-cost drugs prescribed	Numeric (e.g., 8)
F50	Average Number of Services Per Patient	Average number of services provided per patient	Numeric (e.g., 3)
F51	Count for ICD10 Level 1 A00 to B99	Count of diagnoses classified under ICD10 codes A00 to B99	Numeric (e.g., 30)
F52	Count for ICD10 Level 21 Z00 to Z99	Count of diagnoses classified under ICD10 codes Z00 to Z99	Numeric (e.g., 25)
F53	Percentage of Using Unlisted Codes	Percentage of cases where unlisted codes are used	Percentage (e.g., 2%)
F54	Average Number of Claims Per Patient	Average number of insurance claims filed per patient	Numeric (e.g., 1.5)
F55	Max Number of Claims Per Patient	Maximum number of claims filed by a single patient	Numeric (e.g., 5)
F56	Min Number of Claims Per Patient	Minimum number of claims filed by a single patient	Numeric (e.g., 1)
F57	Average Number of Diagnoses Per Claim	Average number of diagnoses per insurance claim	Numeric (e.g., 2)
F58	Max Number of Diagnoses Per Claim	Maximum number of diagnoses per single claim	Numeric (e.g., 4)
F59	Min Number of Diagnoses Per Claim	Minimum number of diagnoses per single claim	Numeric (e.g., 1)
F60	Average Number of Consultations Per Day	Average number of consultations performed by the doctor per day	Numeric (e.g., 5)
F61	Max Number of Consultations Per Day	Maximum number of consultations performed by the doctor in a single day	Numeric (e.g., 10)
F62	Min Number of Consultations Per Day	Minimum number of consultations performed by the doctor in a single day	Numeric (e.g., 1)
F63	Number of Controlled Medicine Drugs Used	Total number of controlled medicines prescribed	Numeric (e.g., 2)
Class	Fraudulent	Indicator of whether the claim is suspected of being fraudulent	Boolean (True/False)

Table 4
Dataset description.

Medical fraud claims	Instances	Features	Classes
Our Dataset	18 000	63	2
Real Data 1 [56]	100 000	23	2
Real Data 2 [57]	1000	34	2

5.1. Dynamic search space adjustment and feature weighting in WOA

For this method, the model includes weighted features and an adaptive search space subroutine. This thus serves to inherently make the most important features more important by giving them a depth of weight. This way the feature that has a larger effect on the model's performance, will be ranked from top to bottom in the optimization process. Then the algorithm adaptively refines the borders of exploration by adjusting the search space dynamically depending on the performance of current solutions. By employing this dynamic mechanism, the algorithm can venture further into the solution space when necessary, and focus on good spots as optimization proceeds. This prevents the model from getting stuck at local optima and thus increases the likelihood that it finds the global optimum.

Furthermore, the features are weighted so that instead of sums we have a vector, and when we multiply this by the value (+1/−1 in our case) of the feature decided by the weighting process. For example, inputting a small data that has two instances, the value of the first feature for each instance will be multiplied by the respective value of the solution vector that is generated in Eq. (8). This guarantees the exact requirements of the dataset. In this way, the importance of each feature is adaptively adjusted by the algorithm such that, the solution aligns with both the structure of data and optimization objectives present in your choice of objective function. Which not only provides a better estimation but is also complementary to the learned parameters from the model.

$$w_{ij} = x_i \times f_{ij} \quad (8)$$

where w_{ij} is the weighted value for the i th feature in the j th instance of the dataset and f_{ij} is the original feature value of the i th feature for the j th instance.

Besides we apply an improvement for WOA by allowing the search space to change during the optimization process. The idea here is that we update the boundaries of search based on the best current solution during optimization. The aim is to dynamically modify the search space by widening or narrowing the boundaries according to how well the algorithm has been searching around them, in order for it actually converge on more promising regions without getting stuck into local optima.

Because of being simplicity and powerful, most computer scientists have been attracted to metaheuristic algorithms for solving real and complex optimization problems, one of which includes WOA. Nevertheless, these algorithms are known to fall into local optima quite easily especially when the search space is static and exploration is limited. In this paper, we propose an adaptive or dynamic Search Space with the Weighted WOA (DW-WOA). Which adjusts the search boundaries on the fly how far they are depending on the current best solution. This method allows the algorithm to explore more which in return increases its chance of finding global optimum.

5.1.1. Mathematical formulation of dynamic search space

Consider a decision variable x with an initial search space defined by the interval $[x_{\min}, x_{\max}]$. At each iteration t of the WOA, we evaluate the current best solution, denoted as x_t^* . If x_t^* is close to the current boundary, we adjust the search space dynamically. The adjustments are expressed mathematically as follows:

- Upper Boundary Adjustment:

$$x_{\max}(t+1) = \begin{cases} x_{\max}(t) + \Delta_x, & \text{if } x_t^* > x_{\max}(t) - \epsilon, \\ x_{\max}(t), & \text{otherwise.} \end{cases} \quad (9)$$

- Lower Boundary Adjustment:

$$x_{\min}(t+1) = \begin{cases} x_{\min}(t) - \Delta_x, & \text{if } x_t^* < x_{\min}(t) + \epsilon, \\ x_{\min}(t), & \text{otherwise.} \end{cases} \quad (10)$$

In these equations:

- ϵ is a small threshold value that determines how close the solution must be to the boundary to trigger an expansion.

- Δ_x represents the expansion step size, which can be a fixed value or dynamically adjusted based on the search progress or convergence speed.

These changes enable the algorithm to investigate other area since we converged in that direction and the best solution x_t^* is near the current limits.

This strategy of course has its own downside, it must implement a balancing act between choosing control parameters that yield optimal search granularity (balance exploration) and answering task-relevant queries (exploitation). If the finite difference Δ_x is too large, the search space may increase unproportionally leading to increased computational cost without much improvement (if any) in solution quality. Inversely, if Δ_x is too little the algorithm will not explore enough far away from local optima.

We also use a dynamic technique for Δ_x varying the expansion factor throughout the optimization process (to cope with these). This can be modeled as:

These adjustments allow the algorithm to explore new regions that were previously outside the initial limits when the best solution x_t^* approaches the current boundaries.

$$\Delta_x(t) = \Delta_x^{\text{initial}} \cdot \left(1 - \frac{t}{T_{\max}}\right)^{\gamma}, \quad (11)$$

where:

- $\Delta_x^{\text{initial}}$ is the initial step size.
- $T_{\max}$ is the maximum number of iterations.
- γ is a decay factor controlling the rate of reduction.

This $\Delta_x(t)$ -derivative-less effect progressively concentrates the search around promising regions and curbs incommensurate (random) expansions with increasing number of iterations. The speed at which $\Delta_x(t)$ is reduced can be controlled by adjusting γ , giving us the flexibility to tune the algorithm's behavior.

For example, consider an initial search space for x grounded at $[0, 10]$ with $\epsilon = 2$, and $\Delta_x^{\text{initial}} = 2$. At iteration t , suppose the best solution = $x_t^* = 9$. Here, we evaluate whether to adjust the upper boundary:

Calculate $x_{\max}(t) - \epsilon$:

$$x_{\max}(t) - \epsilon = 10 - 2 = 8.$$

Since $x_t^* = 9 > 8$, the condition $x_t^* > x_{\max}(t) - \epsilon$ is satisfied. Therefore, we adjust the upper boundary:

$$x_{\max}(t+1) = x_{\max}(t) + \Delta_x(t) = 10 + 2 = 12.$$

Now the search space is widened to $[x_{\min}(t+1), x_{\max}(t+1)] = [0, 12]$, this way it starts exploring solutions other than the initial boundary and increases its chance of finding better solutions.

The second experiment, using The Dynamic Search Space and Weighted Whale Optimization Algorithm (DW-WOA), adjusts the search space based on the current best solution compared to WOA alone. The algorithm adjusts exploration and exploitation balance by tuning the control parameters without a predetermined step size together with an adaptive expansion step.

This dynamic search space strategy makes the DW-WOA realize better exploration of the solution space and then obtains improved results in optimizing over a numerous set of cases. The pseudocode for Dynamic Search Space algorithm is shown in [Algorithm 2](#)

5.2. Fitness assessment

In order to evaluate the individuals generated by the WOA, A fitness function should be considered. In this study, the classification accuracy of the SVM classifier is used as the fitness function for evaluation as shown in Eq. (12). This choice is based on the fact that accuracy directly reflects the model's ability to correctly classify instances in the

Algorithm 2 Dynamic Search Space

```

1: Input: Objective function  $f(x)$ ; initial boundaries  $[x_{\min}, x_{\max}]$ ; pop-
   ulation size  $N$ ; max iterations  $T_{\max}$ ; initial step size  $\Delta x_0$ ; threshold
    $\epsilon$ ; decay factor  $\gamma$ 
2: Initialize whale population  $\{X_i\}_{i=1}^N$  within  $[x_{\min}, x_{\max}]$ 
3: Evaluate fitness  $f(X_i)$  for all whales
4: Set  $x_{\text{best}} = \arg \min f(X_i)$ 
5: for  $t = 1$  to  $T_{\max}$  do
6:   for each whale  $X_i$  do
7:     Update position  $X_i$  using WOA equations
8:     Enforce boundaries:  $X_i \leftarrow \text{clip}(X_i, x_{\min}, x_{\max})$ 
9:   end for
10:  Evaluate fitness  $f(X_i)$ ; update  $x_{\text{best}}$ 
11:  if  $x_{\text{best}} > x_{\max} - \epsilon$  then
12:     $x_{\max} \leftarrow x_{\max} + \Delta x(t)$ 
13:  end if
14:  if  $x_{\text{best}} < x_{\min} + \epsilon$  then
15:     $x_{\min} \leftarrow x_{\min} - \Delta x(t)$ 
16:  end if
17:  Update step size:  $\Delta x(t) = \Delta x_0 \left(1 - \frac{t}{T_{\max}}\right)^{\gamma}$ 
18: end for
19: Return best solution  $x_{\text{best}}$ 

```

dataset. Also, the algorithm can effectively assess the quality of each solution, guiding the search towards models that offer better predictive performance.

$$\text{fitness}(I_t^i) = \frac{1}{V} \sum_{v=1}^V \frac{1}{N} \sum_{j=1}^N \delta(c(x_j), y_j) \quad (12)$$

where the $c(x_j)$ is the accuracy for the instance j th, and y_j is the actual class label for that instance. To relationship between y_j and $c(x_j)$ we used δ . So, if $\delta = 1$, then $c(x_j)$ equals y_j , otherwise $\delta = 0$. Furthermore, N denotes the total number of instances, while v and V represent the current fold and the total number of folds in the cross-validation process, respectively.

5.3. Procedure of DW-WOA-SVM

The process of DW-WOA-SVM starts by splitting the datasets into training and testing sets. The splitting criteria are determined according to the predefined number of runs. In our case, it is divided into k folds for each run or experiment conducted, where $k - (1/k)$ is the training set and the remaining $1/k$ is the testing set. This ensures the diversity of the data as much as possible, resulting in the most robust model.

After that, the WOA began its operation by randomly generating a vector of real numbers that will be used to set up the parameters C and γ , and the weights of the features. Then, the SVM commences the training process using the weighted training set. During this process, an inner cross-validation is performed to build a robust model that can prevent overfitting. It is worth noting that the number of folds in the inner cross-validation is 3. Specifically, the process works as follows:

1. **Parameter Evaluation:** Each candidate solution from WOA is evaluated using inner CV for its SVM parameters (C , γ) and feature weights.
2. **K-Fold Validation:** The training set is divided into k -folds (e.g., $k = 3$), with $k - 1$ folds for training and the remaining fold for validation, repeated k times.
3. **Fitness Calculation:** The average accuracy across k -folds serves as the fitness score, preventing overfitting to specific training subsets.
4. **WOA Integration:** The fitness score refines WOA's search for optimal parameters and feature weights.

Table 5
Confusion matrix.

	Predicted positive	Predicted negative
Actual Positive	TP	FN
Actual Negative	FP	TN

When the training phase is done, the SVM sends the fitness function to the WOA, which can subsequently be used to enhance fitness (accuracy). This operation is repeated on the training set until the termination condition of the WOA is met. In this instance, the termination criteria are the maximum number of iterations. Furthermore, once the maximum number of WOA generations has been reached, the best solution will be applied to evaluate the model using the testing set. Finally, all the steps are repeated according to the specified number of runs, and the average results will be taken into account.

5.4. Evaluation

This subsection describe the evaluation measures used in the work. In order to assessments of the models we used the confusion matrix [Table 5](#). The measurements used in this study are accuracy, recall, precision, and f-measure. Based on this assumption, each evaluation measure is presented as follows:

- **Accuracy:** The number of fraud and non-fraud instances that were accurately classified is divided by the total number of instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

- **Precision:** The percentage of fraud that have been classified correctly over all the expected fraud instances.

$$Precision = \frac{TP}{TP + FP} \quad (14)$$

- **Recall:** measured by taking the number of correctly anticipated fraud divided by the total number of fraud.

$$Recall = \frac{TP}{TP + FN} \quad (15)$$

- **F-measure :** its the harmonized mean of precision and recall, which can be calculated using the following equation:

$$F - measure = \frac{2 * (precision * recall)}{precision + recall} \quad (16)$$

The explanation of all the phases presented in [Fig. 5](#).

6. Experiments and results

This section presents the results of the experiments conducted on the proposed algorithm, alongside a comparison with other established algorithms. The analysis is divided into several phases. The first phase is composed of two parts: the initial part evaluates the performance of standard classifiers (J48, Random Forest, Naive Bayes, AdaBoost, Bagging, and Support Vector Machine) across three datasets — two real-world datasets and one synthetically generated dataset. The objective is to assess how the generated data compares with the real datasets in terms of classification performance. The second part of this phase involves applying a filter-based feature selection method to examine its impact on the classifiers' performance.

In the second phase, the best-performing classification method from the first phase, SVM, is combined with several metaheuristic algorithms for various tasks. This phase is divided into three parts. The first part involves using metaheuristic algorithms solely for parameter optimization without feature selection. The second part applies metaheuristic algorithms for both parameter optimization and feature selection. In

Table 6
The detailed settings of the used system.

Name	Settings
Hardware	
CPU	AMD Ryzen 5 5600X processor
Frequency	3.7 GHz
# of Cores	6
RAM	16 GB
Hard drive	1 TB
Software	
Operating system	Windows 10 (64-bit)
Application	Matlab

Table 7
A description of the initial parameters used for the metaheuristic algorithm.

Algorithm	Parameter	Value
GA [58]	Crossover ratio	0.9
	Mutation ratio	0.1
MVO [59]	Minimum wormhole existence ratio	0.2
	Maximum wormhole existence ratio	1
PSO [60]	Acceleration constants	[2.1, 2.1]
	Inertia w	[0.9, 0.6]
WOA [55]	a	2 to 0
	r	[0, 1]
SSA [61]	c_1	[0-1]
	c_2	[0-1]
HHO [62]	Initial energy (E_0)	[-1,1]
	Jump strength (J)	[0, 2]

the third part, the proposed approach is integrated with several metaheuristic algorithms to perform weighted feature selection and dynamic search space optimization.

In the third phase, we compared the proposed approach against the weighted approach alone for more examination. The experiments were conducted on 14 well-known benchmark datasets.

6.1. Experiments setup

To ensure a fair comparison, all experiments in this study were conducted under consistent conditions. [Table 6](#) details the specifications of the personal computer we used for our experiments. We further validated the training and testing sets through 10-fold cross-validation to enhance the reliability of our results. This comprehensive setup allows us to thoroughly assess the performance of our approach across different datasets and benchmarks. Moreover, [Table 7](#) presents the initial parameters of the metaheuristic algorithms utilized in this study.

6.2. Phase one: Standard classifiers

This subsection outlines the performance results of the standard classifiers on the three datasets and is divided into two parts: the first focuses on the classical classification methods applied independently, while the second examines the classifiers combined with filter-based feature selection.

6.2.1. Part 1: Results of standard classification methods

[Table 8](#) presents the results of various standard classification methods applied to Real Data 1, in terms of accuracy, recall, precision, and F-measure. Among all the algorithms, the best accuracy was achieved by SVM with 82.2%, followed by J48 with 81.1%. Additionally, J48 obtained the highest recall and F-measure, with 67.6% and 63.9% respectively, followed by SVM with 60.7% and 62.8%. In terms of precision, NB achieved the best result with 64.9%. The high accuracy

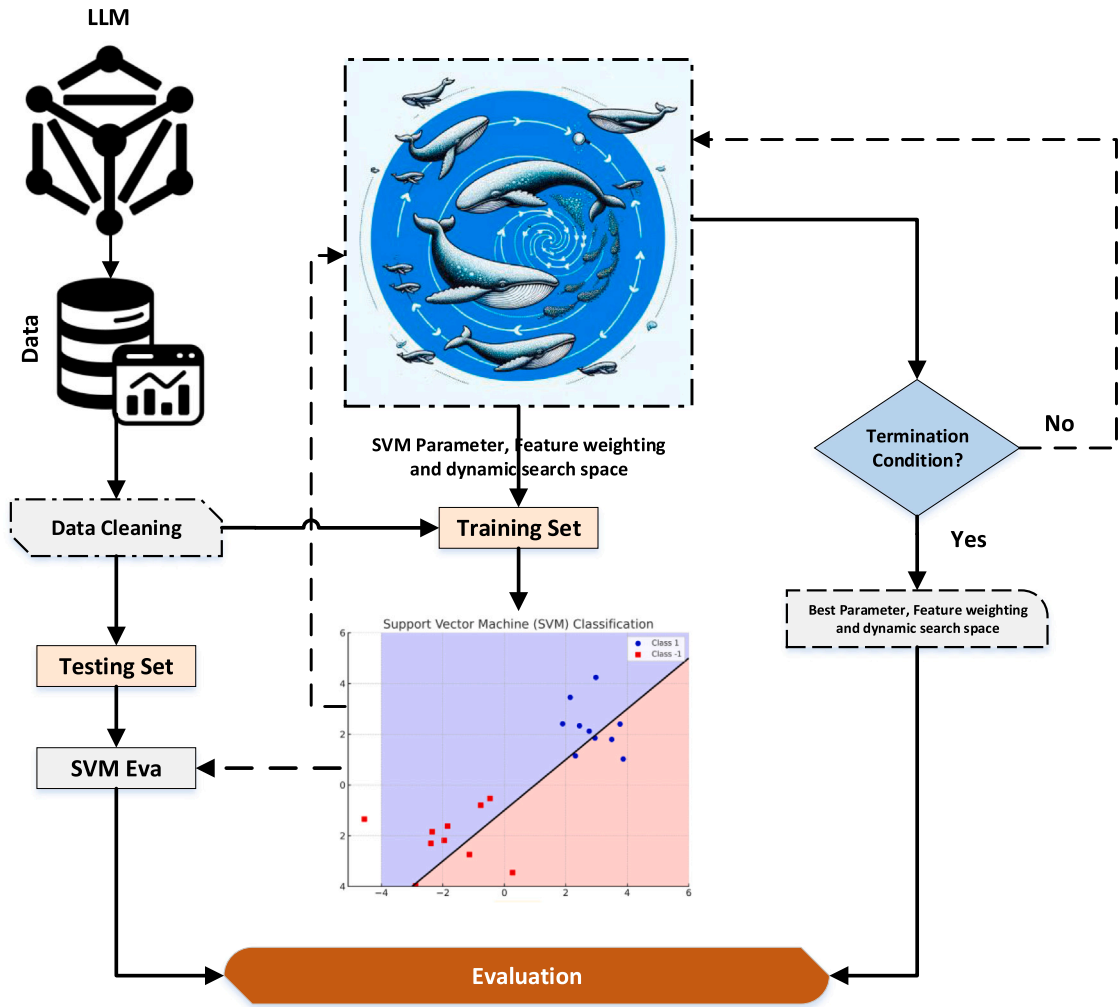


Fig. 5. The proposed DW-WOA-SVM process.

Table 8
Results of the standard classification method on the Real dataset 1.

	Accuracy	Recall	Precision	F-measure
J48	0.811	0.676	0.605	0.639
RF	0.747	0.150	0.463	0.226
NB	0.736	0.478	0.478	0.585
AdaBoost	0.781	0.453	0.571	0.506
Bagging	0.753	–	–	–
SVM	0.822	0.607	0.649	0.628

of SVM indicates its effectiveness in correctly classifying the majority of instances in the dataset.

As for Table 9, the results show that, similar to the previous experiment, the SVM achieved the highest accuracy at 76.7%. In terms of recall, the SVM also attained the best result with 100%. Regarding precision and F-measure, NB and J48 achieved the best values, with 78.4% and 96.7%, respectively. Therefore, these results highlight the consistent performance of SVM in terms of accuracy and recall, while NB and J48 demonstrate their strength in precision and F-measure, indicating a well-rounded model evaluation across different metrics.

Table 10 presents the results for our generated data and its performance. As in the two previous experiments, the SVM achieved the highest accuracy at 81.5%. The Bagging classifier achieved the best result in terms of recall, with 100%, while NB attained the best precision with 92.4%. Regarding the F-measure, the best value was

Table 9
Results of the standard classification method on the Real dataset 2.

	Accuracy	Recall	Precision	F-measure
J48	0.750	0.967	0.768	0.967
RF	0.761	0.991	0.766	0.864
NB	0.639	0.731	0.784	0.757
AdaBoost	0.762	0.992	0.767	0.865
Bagging	0.763	0.992	0.767	0.865
SVM	0.767	1.000	0.767	0.868

Table 10
Results of the standard classification method on the generated dataset.

	Accuracy	Recall	Precision	F-measure
J48	0.809	0.922	0.815	0.865
RF	0.813	0.936	0.811	0.869
NB	0.811	0.779	0.924	0.845
AdaBoost	0.808	0.785	0.913	0.844
Bagging	0.661	1.000	0.662	0.797
SVM	0.815	0.908	0.830	0.867

achieved by RF, with 86.9%. These results demonstrate that our generated data closely mirrors the characteristics of the real datasets, leading to comparable performance across classifiers. This indicates that the synthetic data effectively captures the underlying patterns, allowing the models to generalize similarly to how they perform on real-world data.

Table 11

Results of the standard classification method combined with filter feature selection on the generated dataset.

	Accuracy	Recall	Precision	F-measure
J48	0.811	0.810	0.896	0.851
RF	0.661	1.000	0.662	0.797
NB	0.810	0.779	0.923	0.845
AdaBoost	0.810	0.779	0.923	0.845
Bagging	0.808	0.781	0.918	0.844
SVM	0.832	0.814	0.895	0.853

6.2.2. Part 2: Results of the filter feature selection method

In the second part of phase one, we perform the filter feature selection method to investigate and examine the performance of the standard classification methods on the newly generated datasets. It is worth noting that filter feature selection is a technique used to identify and select relevant features by applying statistical measures, such as correlation or information gain, before the learning process begins. This approach enhances model performance by reducing dimensionality and minimizing the risk of overfitting, while maintaining a more interpretable feature set.

Table 11 illustrates the results after the impact of the filter feature selection method. The SVM here also achieved the best results with 83.2% with a rise of 1.7%. As for recall and precision measures, the RF obtained the best results with 100% in recall, while NB, and AdaBoost both achieved 92.3%. Additionally, the SVM accomplished the highest results in terms of f-measure with 86.9%. We notice that the results only rise in accuracy; this is due to the filter feature selection method removing irrelevant or redundant features, allowing the classifier to focus on the most informative attributes. Consequently, the improved feature set enhances the model's performance, highlighting the effectiveness of feature selection in classification tasks.

Moreover, Fig. 6 lists the best-selected features identified by the filter method. In the case of medical insurance fraud detection, the selection of features with filter feature selection is pivotal, as these features offer profound insights into potential discrepancies. Features such as the *Average Number of Patients Per Week* and *Average Number of Consultations Per Day* are critical; they illuminate patterns of service volume that may surpass typical ranges, possibly indicating inflated claims or fictitious billing scenarios. Similarly, the *Policy Number* and *Claim Description* provide a window into policy-specific behaviors and claim irregularities, revealing abuse patterns that warrant closer scrutiny. Dates like the *Settlement Date* and *Policy Start Date* also play a significant role by shedding light on the timing and frequency of claims, which are essential for uncovering premeditated fraudulent activities within the healthcare system.

In summary, the evaluation demonstrates that SVM consistently achieves the highest accuracy across all datasets, including our synthetic data, highlighting its robustness in classification tasks. The incorporation of filter-based feature selection further enhances model performance by removing irrelevant features, leading to improved accuracy and more focused learning. These findings underscore the effectiveness of combining SVM with feature selection techniques to achieve superior classification results on both real and generated datasets.

6.3. Phase two: Results of metaheuristic algorithms

In this phase, we combined various metaheuristic algorithms (GA, HHO, MVO, PSO, SSA, and WOA) with the best classifier from the previous experiments, namely SVM. This phase consists of three parts: Without Feature Selection, With Feature Selection, and the Dynamic Weighted Search Space Approach. It is worth noting that the experiments in this phase will be conducted exclusively on our generated dataset.

6.3.1. Part 1: Metaheuristic algorithms without feature selection (WFS)

The first part describes the combination of SVM with different metaheuristic algorithms to optimize its parameters without performing feature selection. Table 12 illustrates the results of six approaches, namely, GA-SVM-WFS, HHO-SVM-WFS, MVO-SVM-WFS, PSO-SVM-WFS, SSA-SVM-WFS, and WOA-SVM-WFS, in terms of accuracy, recall, precision, and F-measure. As can be seen in the table, the results show the best performance for each measure. For instance, the best accuracy was obtained by WOA-SVM-WFS with 86.916%, followed by SSA-SVM-WFS, HHO-SVM-WFS, MVO-SVM-WFS, GA-SVM-WFS, and PSO-SVM-WFS, respectively. As for the other measures, recall and F-measure also achieved the highest results with WOA-SVM-WFS, obtaining 87.098% and 92.150%, respectively. In terms of precision, GA-SVM-WFS achieved the best result with 100%.

Therefore, WOA-SVM-WFS emerges as the most effective approach for optimizing SVM parameters without feature selection on our generated dataset, achieving the highest scores in accuracy, recall, and F-measure. However, GA-SVM-WFS attained the highest precision, indicating its potential usefulness in applications where precision is particularly important.

Furthermore, it is worth mentioning that this experiment achieves a 3.7% improvement over the previous one, highlighting the effectiveness of using metaheuristic algorithms for parameter optimization. Moreover, Table 13 displays the p-values from the Wilcoxon statistical test, assessing the performance of WOA-SVM-WFS and other algorithms on the generated data. The results reveal statistically significant differences (p less 0.05) across all algorithms.

6.3.2. Part 2: Metaheuristic algorithms with feature selection (FS)

The second part depicts the performance of using parameter optimization and feature selection simultaneously. In this part, Table 14 presents the results of the six approaches while performing parameter optimization and feature selection: GA-SVM-FS, HHO-SVM-FS, MVO-SVM-FS, PSO-SVM-FS, SSA-SVM-FS, and WOA-SVM-FS.

The results show that WOA-SVM-FS achieved the best results in terms of accuracy with 88.460%, followed by HHO-SVM-FS, MVO-SVM-FS, PSO-SVM-FS, SSA-SVM-FS, and GA-SVM-FS, respectively. Also, the same approach achieved the best results in terms of recall with 98.789%. As for the other measures, precision, and F-measure, HHO-SVM-FS acquired the highest values with 95.559% and 91.686%, respectively.

Compared to the previous results (without feature selection), the feature selection approach improved the accuracy by 1.544%. Therefore, the simultaneous use of parameter optimization and feature selection leads to a noticeable improvement in classifier performance. Further, Table 15 displays the p-values from the Wilcoxon statistical test, assessing the performance of WOA-SVM-WFS and other algorithms on the generated data. The results indicate statistically significant differences for all algorithms except HHO.

6.3.3. Part 3: Dynamic weighted search space approach using metaheuristic algorithm

The third part describes the performance of our proposed approach (Dynamic Weighted Search Space approach) on the generated dataset. In other words, the approach consists of applying both the weighted features and dynamic search space methods at the same time.

Table 16 illustrates the results of the proposed approach on different algorithms, namely DW-GA-SVM, DW-HHO-SVM, DW-MVO-SVM, DW-PSO-SVM, DW-SSA-SVM, and DW-WOA-SVM. As in the previous two experiments, the DW-WOA-SVM also obtained the best results in terms of accuracy with 99.199%, followed by DW-GA-SVM, DW-SSA-SVM, DW-HHO-SVM, DW-MVO-SVM, and DW-PSO-SVM, respectively. The DW-WOA-SVM also achieved the best results in recall and F-measure with 99.341% and 99.386%, respectively, while the best precision was obtained by DW-PSO-SVM.

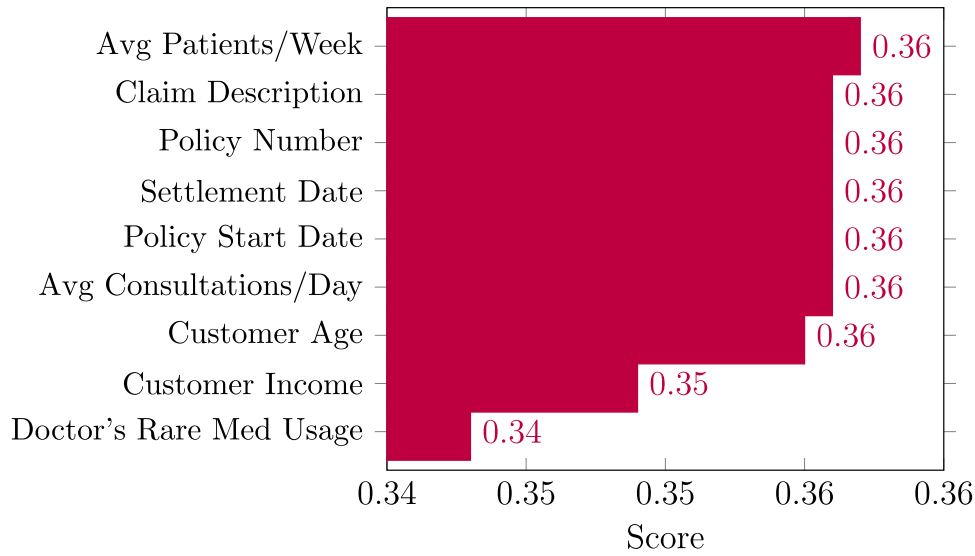


Fig. 6. Feature importance scores using filter feature selection.

Table 12

Results of different metaheuristic algorithms combined with SVM, without feature selection, on the generated dataset.

Algorithm	Accuracy		Recall		Precision		F-measure	
	Avg	Std	Avg	Std	Avg	Std	Avg	Std
GA-SVM-WFS	73.119	14.235	73.061	14.120	100.000	0.000	83.802	19.851
HHO-SVM-WFS	82.320	18.138	82.353	18.174	99.841	0.335	89.253	10.970
MVO-SVM-WFS	76.060	15.804	76.434	16.400	99.436	0.923	85.530	29.452
PSO-SVM-WFS	69.559	10.586	69.584	10.663	99.918	0.259	81.655	9.846
SSA-SVM-WFS	82.567	17.446	82.726	17.616	99.595	0.687	89.418	10.458
WOA-SVM-WFS	86.916	16.023	87.098	16.178	99.591	0.576	92.150	9.474

Table 13

P-values from the Wilcoxon statistical test comparing WOA-SVM-WFS and other algorithms on the generated dataset.

Algorithms	P-value
GA	8.15E-09
HHO	0.0136
MVO	6.68E-12
PSO	2.82E-19
SSA	5.80E-04

Compared to the previous experiments, the proposed approach (DW-WOA-SVM) gained an increase of 10.73% and 12.28% compared to WOA-SVM-FS and WOA-SVM-WFS, respectively. This proves the superiority of our approach over the approaches without and with feature selection.

Overall, these results demonstrate that our proposed Dynamic Weighted Search Space approach significantly outperforms both the parameter optimization without feature selection and with feature selection methods. This highlights the effectiveness of combining weighted features with a dynamic search space in enhancing SVM classifier performance on our generated dataset. Further, Table 17 displays the p-values from the Wilcoxon statistical test, assessing the performance of DW-WOA-SVM and other algorithms on the generated data. The results reveal statistically significant differences across all algorithms.

6.3.4. Feature importance analysis

The proposed approach weighted the best feature that affects and helps in the prediction of the fraud claims performance. As shown in Fig. 7, the top 20 features are listed, where the best feature was *Total*

Number of Patients Per 1st Weekend in the Month and the worst was *Percentage of Using Unlisted Codes*.

Further analysis was conducted to study these features, focusing specifically on the top 5 features. For example, the top feature, *Total Number of Patients Per 1st Weekend in the Month*, could indicate higher activity periods that may be associated with abnormal claim behaviors or fraud trends, as weekends might exhibit different patterns compared to weekdays. The second feature (*Min Number of Claims Per Patient*) shows that a low number of claims might be typical for most patients, but deviations can indicate either lack of necessary care or potential fraudulent behavior, such as underutilization for cost-saving or overutilization through falsified claims.

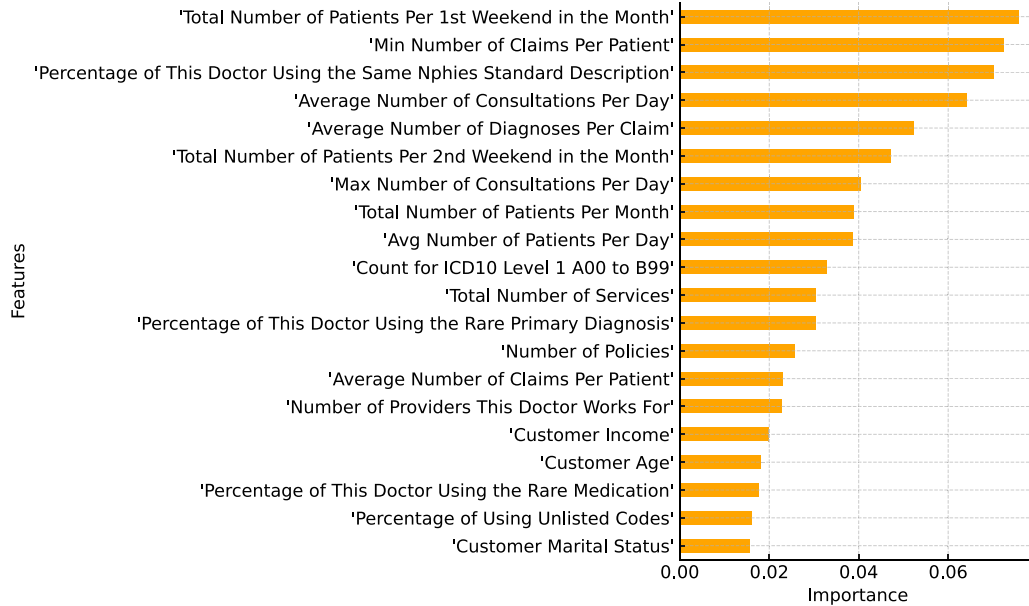
The third feature, *Percentage of This Doctor Using the Same Nphies Standard Description*, indicates the consistency in diagnoses or procedures used in medical descriptions by doctors. High uniformity could suggest a templated approach to filing claims, which may be a potential indicator of fraudulent behavior. As for the fourth feature, *Average Number of Consultations Per Day*, it suggests identifying outliers in the daily operations of medical providers. A high number of daily consultations might indicate rushed, artificially inflated claims or inadequate care. Whereas the fifth feature, *Average Number of Diagnoses Per Claim*, demonstrates that an unusually high number of diagnoses per claim may suggest the addition of unnecessary diagnoses to increase the claim amount.

In conclusion, this in-depth analysis offers valuable insights into identifying potential fraudulent behavior within healthcare claims. By examining patterns such as claim frequency, provider consistency, and operational anomalies, the study highlights key indicators that may suggest irregularities or manipulation. These findings underscore the importance of targeted feature analysis in enhancing the accuracy and effectiveness of fraud detection systems, ultimately contributing to the integrity of healthcare processes and ensuring more reliable outcomes.

Table 14

Results of different metaheuristic algorithms combined with SVM, with feature selection, on the generated dataset.

Algorithm	Accuracy		Recall		Precision		F-measure	
	Avg	Std	Avg	Std	Avg	Std	Avg	Std
GA-SVM-FS	80.554	3.673	81.590	4.196	91.554	5.493	86.125	66.891
HHO-SVM-FS	88.258	9.586	88.352	9.769	95.559	3.506	91.686	79.747
MVO-SVM-FS	86.652	9.003	87.451	8.607	94.329	5.361	90.619	76.519
PSO-SVM-FS	86.428	9.257	86.645	9.630	94.921	4.573	90.390	76.662
SSA-SVM-FS	83.111	6.785	84.449	7.681	91.683	7.852	87.576	72.143
WOA-SVM-FS	88.460	9.486	89.789	9.362	94.078	6.968	91.640	81.067

**Fig. 7.** The most important features based on the prospered approach.**Table 15**

P-values from the Wilcoxon statistical test comparing WOA-SVM-FS and other algorithms on the generated dataset.

Algorithm	P-value
GA	2.63E-07
HHO	0.0504
MVO	0.024
PSO	1.29E-04
SSA	9.37E-04

6.4. Phase 3: Proposed approach vs weighted approach

In this phase, further investigation was conducted to examine the proposed approach against the standalone weighted approach. Table 16 presents the results of the comparison between the proposed approach and the weighted approach on 14 benchmarks. As shown in the table, the proposed DW-WOA-SVM outperformed the W-WOA-SVM approach in 10 out of 14 datasets. This proves the effectiveness of our approach against different well-known benchmarks. In more detail, the DW-WOA-SVM achieved better results in all datasets except the spectftf, vowel, diabetes, and libras datasets.

Furthermore, the proposed DW-WOA-SVM approach demonstrates superior performance over the standalone weighted approach, outperforming it in 10 out of 14 benchmark datasets. This indicates the robustness and effectiveness of our method across a variety of well-known benchmarks. Therefore, the DW-WOA-SVM approach offers a significant improvement for SVM classification tasks in diverse applications (see Table 18).

Moreover, the convergence curve of the results is shown in Fig. 8, providing a clear visualization of how quickly and efficiently each approach converges towards optimal solutions. As observed, the proposed Dynamic Weighted Search Space approach achieved the best convergence in 8 datasets, demonstrating faster and more stable optimization. It performed equally well (a tie) in 3 datasets, indicating comparable convergence speed to the weighted approach, and showed the slowest convergence in 3 datasets. This suggests that while the proposed approach is generally superior in terms of convergence, there are a few cases where the weighted approach may be equally or slightly more efficient. Overall, the convergence results further reinforce the advantages of the DW-WOA-SVM approach across most benchmarks.

Table 19 presents the p-values obtained from the Wilcoxon statistical test, comparing the performance of DW-WOA-SVM and W-WOA-SVM over 14 datasets. The results indicate statistically significant differences (p less than 0.05) for most datasets, except for “diabetes” and “glass”, where the differences were not significant.

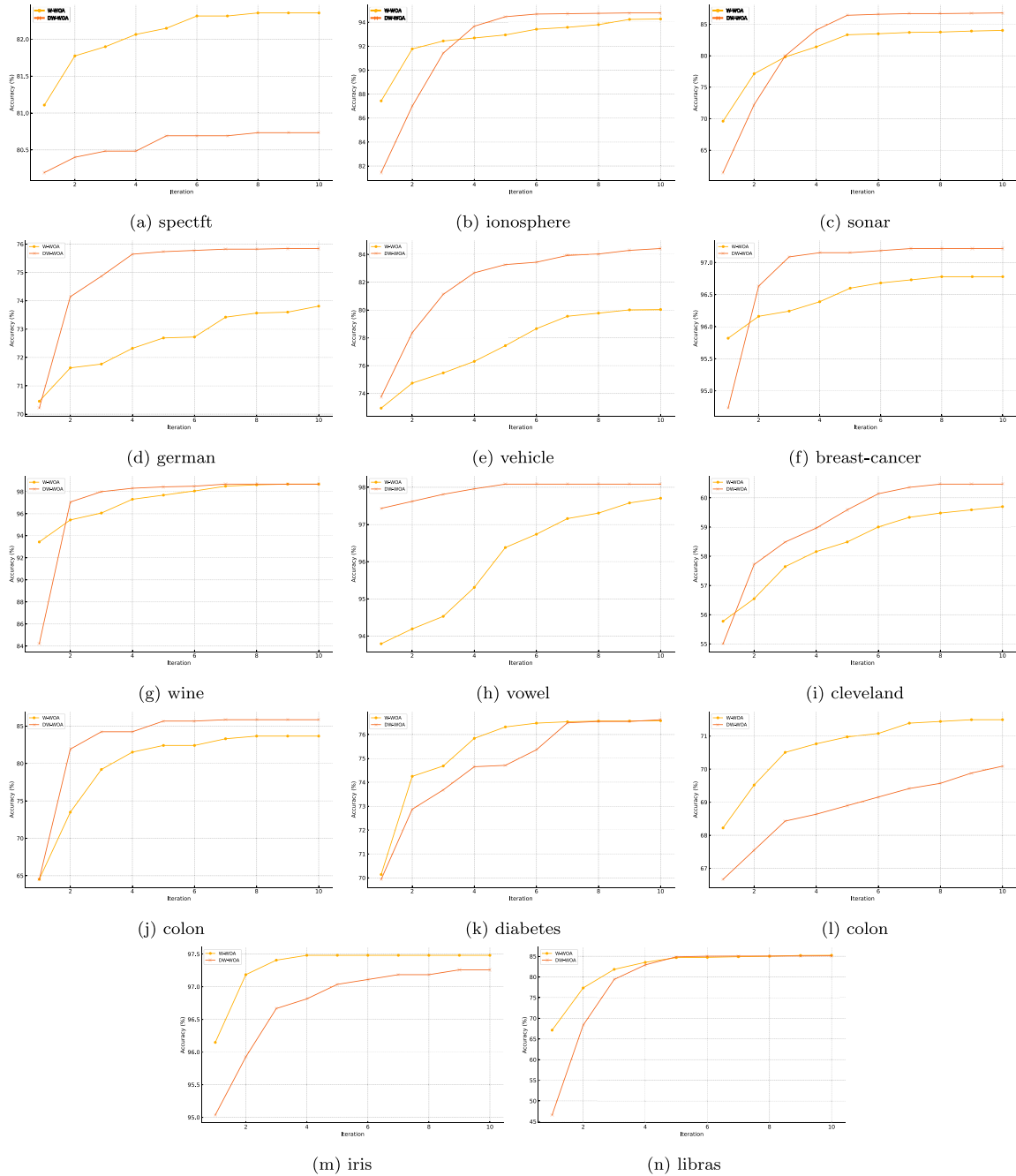
7. Conclusion

Fraud in medical claims causes numerous problems in healthcare systems, hindering their ability to effectively help patients. Therefore, it is crucial to identify these fraudulent activities to allow healthcare systems to continue improving. In the literature, researchers have attempted to mitigate this problem; however, with the evolution of fraud schemes, detecting such activities has become increasingly difficult. In this work, we propose a novel approach named the evolutionary dynamic weighted search space approach (DW-WOA-SVM). This approach uses optimization algorithms to improve parameters, weight features,

Table 16

Results of the proposed Dynamic Weighted Search Space approach on the generated dataset.

Algorithm	Accuracy		Recall		Precision		F-measure	
	Avg	Std	Avg	Std	Avg	Std	Avg	Std
DW-GA-SVM	98.398	1.127	98.603	1.050	99.033	1.413	98.810	0.820
DW-HHO-SVM	89.001	17.012	88.976	17.003	99.843	0.332	93.243	10.381
DW-MVO-SVM	86.916	16.426	86.937	16.447	99.835	0.349	92.150	9.782
DW-PSO-SVM	75.207	16.927	75.207	16.927	99.920	0.253	84.908	10.311
DW-SSA-SVM	96.311	9.621	96.422	9.656	99.587	0.585	97.719	5.667
DW-WOA-SVM	99.199	0.454	99.341	0.673	99.436	0.545	99.386	0.369

**Fig. 8.** Convergence curves of the proposed approaches for KrvspeEW, Tic-tac-toe, Vote, WaveformEW, WineEW, Zoo, Clean1, Semeion, Colon, and Leukemia datasets.

and perform dynamic search space optimization simultaneously. Additionally, LLMs have been used to generate synthetic data that resembles real datasets but with more features to enhance the richness of the data.

The experiments in this study consist of three phases, each focusing on a specific aspect: standard classification methods, the metaheuristic algorithms phase, and the benchmarks phase. The proposed approach

Table 17

P-values from the Wilcoxon statistical test comparing WOA-SVM-FS and other algorithms on the generated dataset.

Algorithm	P-value
GA	3.62E−09
HHO	0.0457
MVO	0.0355
PSO	7.00E−15
SSA	0.0112

Table 18

Results of the proposed Dynamic Weighted Search Space approach compared to the weighted approach on various well-known benchmarks.

Dataset	DW-WOA-SVM		W-WOA-SVM	
	Accuracy		Accuracy	
	Avg	Std	Avg	Std
spectft	76.809	7.678	77.165	5.308
ionosphere	94.310	3.790	91.159	5.305
sonar	89.452	5.821	84.500	9.876
german	75.800	2.530	73.500	5.061
vehicle	84.864	3.640	80.482	4.908
breast_cancer	96.628	2.202	95.906	2.548
wine	97.222	3.928	95.458	3.667
vowel	98.864	1.825	99.623	0.796
Cleveland	57.161	11.197	55.161	15.014
colon_cancer	84.048	13.043	77.143	14.339
diabetes	74.217	3.773	75.391	4.180
glass	69.134	12.584	67.684	11.463
iris	94.000	7.337	95.333	8.344
libras	88.056	7.297	87.778	3.750

Table 19

P-values from the Wilcoxon statistical test comparing DW-WOA-SVM and W-WOA-SVM across multiple datasets.

Data	P-value
spectft	5.76E−15
ionosphere	9.32E−24
sonar	1.41E−19
german	1.20E−31
vehicle	9.69E−34
breast_cancer	1.73E−20
wine	3.83E−24
vowel	7.79E−34
cleveland	1.54E−33
colon_cancer	1.40E−33
diabetes	0.5177
glass	0.6724
iris	0.0125
libras	2.22E−33

demonstrated its superiority compared to other methods, achieving 99.199% accuracy. In the benchmark phase, it obtained the best results in 10 out of 14 datasets. Furthermore, an analysis of the features and their importance in identifying fraud in medical claims has been discussed.

As for future work, we are thinking of expanding the dataset to further validate its effectiveness, also, explore the potential of applying the blockchain technology for secure and transparent data. Moreover, incorporating additional feature selection techniques.

CRedit authorship contribution statement

Mohammad Tubishat: Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization.

Dina Tbaishat: Supervision, Funding acquisition, Conceptualization. **Ala' M. Al-Zoubi:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Abed-Elalim Hraiz:** Visualization, Software, Data curation. **Maria Habib:** Writing – review & editing, Writing – original draft, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] E. Nabrawi, A. Alanazi, Fraud detection in healthcare insurance claims using machine learning, *Risks* 11 (9) (2023) 160.
- [2] K. Doheny, Montana hospital to pay 10.8m to settle false claims oncology suit, 2024, URL: <https://www.medscape.com/viewarticle/montana-hospital-pay-10-8m-settle-false-claims-oncology-suit-2024a1000glj?form=fpf>.
- [3] H. Rabinowitz, Justice department charges nearly 200 people in 2.7 billion health care fraud crackdown, 2024, URL: <https://edition.cnn.com/2024/06/27/us/health-care-fraud-crackdown/index.html/>.
- [4] L. Victory, Nationwide medicare scams have taken 100m meant for senior's medical bills, investigation reveals, 2024, URL: <https://www.cbsnews.com/chicago/news/illinois-medicare-fraud/>.
- [5] I.D. Mienye, N. Jere, Deep learning for credit card fraud detection: A review of algorithms, challenges, and solutions, *IEEE Access* (2024).
- [6] H.A. Javaid, Improving fraud detection and risk assessment in financial service using predictive analytics and data mining, *Integr. J. Sci. Technol.* 1 (8) (2024).
- [7] A.-Z. Ala'M, A.M. Mora, H. Faris, A multilingual spam reviews detection based on pre-trained word embedding and weighted swarm support vector machines, *IEEE Access* (2023).
- [8] O.A. Alzubi, J.A. Alzubi, I. Qiqieh, A. Al-Zoubi, An IoT intrusion detection approach based on salp swarm and artificial neural network, *Int. J. Netw. Manage.* (2024) e2296.
- [9] T. Kanan, A. Aldaaja, B. Hawashin, Cyber-bullying and cyber-harassment detection using supervised machine learning techniques in arabic social media contents, *J. Internet Technol.* 21 (5) (2020) 1409–1421.
- [10] T. Kanan, A. AbedAlghafer, S. AlZu'bi, B. Hawashin, A. Mughaid, G. Kanaan, M. Kamruzzaman, An intelligent health care system for detecting drug abuse in social media platforms based on low resource language, *IEEE/ACM Trans. Audio, Speech, Lang. Process.* (2023).
- [11] R. Zhang, D. Cheng, J. Yang, Y. Ouyang, X. Wu, Y. Zheng, C. Jiang, Pre-trained online contrastive learning for insurance fraud detection, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, (no. 20) 2024, pp. 22511–22519.
- [12] B. Hong, P. Lu, H. Xu, J. Lu, K. Lin, F. Yang, Health insurance fraud detection based on multi-channel heterogeneous graph structure learning, *Heliyon* 10 (9) (2024).
- [13] S. AlZu'bi, A. Zreiqat, W. Radi, A. Mughaid, L. Abualigah, An intelligent healthcare monitoring system-based novel deep learning approach for detecting covid-19 from x-rays images, *Multimedia Tools Appl.* (2024) 1–18.
- [14] R.H. Chowdhury, Advancing fraud detection through deep learning: A comprehensive review, *World J. Adv. Eng. Technol. Sci.* 12 (2) (2024) 606–613.
- [15] Y. Shang, S. Wang, N. Tang, Y. Fu, K. Wang, Research progress in fault detection of battery systems: A review, *J. Energy Storage* 98 (2024) 113079.
- [16] J. Liu, G. Xie, J. Wang, S. Li, C. Wang, F. Zheng, Y. Jin, Deep industrial image anomaly detection: A survey, *Mach. Intell. Res.* 21 (1) (2024) 104–135.
- [17] D.A. Al-Qudah, A.-Z. Ala'M, A.I. Cristea, J.J. Merelo-Guervós, P.A. Castillo, H. Faris, Prediction of sentiment polarity in restaurant reviews using an ordinal regression approach based on evolutionary XGBoost, *PeerJ Comput. Sci.* 11 (2025) e2370.
- [18] S. Kumar, G.G. Tejani, N. Pholdee, S. Bureerat, P. Mehta, Hybrid heat transfer search and passing vehicle search optimizer for multi-objective structural optimization, *Knowl.-Based Syst.* 212 (2021) 106556.

- [19] S. Kumar, P. Jangir, G.G. Tejani, M. Premkumar, A decomposition based multi-objective heat transfer search algorithm for structure optimization, *Knowl.-Based Syst.* 253 (2022) 109591.
- [20] S. Kumar, N. Panagant, G.G. Tejani, N. Pholdee, S. Bureerat, N. Mashru, P. Patel, A two-archive multi-objective multi-verse optimizer for truss design, *Knowl.-Based Syst.* 270 (2023) 110529.
- [21] S. Kumar, G.G. Tejani, P. Mehta, S.M. Sait, A.R. Yildiz, S. Mirjalili, Optimization of truss structures using multi-objective cheetah optimizer, *Mech. Based Des. Struct. Mach.* (2024) 1–22.
- [22] R. Obiedat, L. Al-Qaisi, R. Qaddoura, O. Harfoushi, A. Al-Zoubi, An intelligent hybrid sentiment analyzer for personal protective medical equipments based on word embedding technique: The covid-19 era, *Symmetry* 13 (12) (2021) 2287.
- [23] B.I. Al-Ahmad, A.-Z. Ala'A, M.F. Kabir, M. Al-Tawil, I. Aljarah, Swarm intelligence-based model for improving prediction performance of low-expectation teams in educational software engineering projects, *PeerJ Comput. Sci.* 8 (2022) e857.
- [24] R.A. Sowah, M. Kuuboore, A. Ofoli, S. Kwofie, L. Asiedu, K.M. Koumadi, K.O. Apeadu, Decision support system (DSS) for fraud detection in health insurance claims using genetic support vector machines (GSVMs), *J. Eng.* 2019 (1) (2019) 1432597.
- [25] S. Anam, M.R.A. Putra, Z. Fitriah, I. Yanti, N. Hidayat, D.M. Mahanani, Health claim insurance prediction using support vector machine with particle swarm optimization, *BAREKENG: J. Ilmu Mat. Dan Ter.* 17 (2) (2023) 0797–0806.
- [26] E. Kasneci, K. Seßler, S. Küchemann, M. Bannert, D. Dementieva, F. Fischer, U. Gasser, G. Groh, S. Günnemann, E. Hüllermeier, et al., ChatGPT for good? On opportunities and challenges of large language models for education, *Learn. Individ. Differ.* 103 (2023) 102274.
- [27] B. Adler, N. Agarwal, A. Aithal, D.H. Anh, P. Bhattacharya, A. Brundyn, J. Casper, B. Catanzaro, S. Clay, J. Cohen, et al., Nemotron-4 340b technical report, 2024, arXiv preprint [arXiv:2406.11704](https://arxiv.org/abs/2406.11704).
- [28] P. Maini, S. Seto, H. Bai, D. Grangier, Y. Zhang, N. Jaitly, Rephrasing the web: A recipe for compute and data-efficient language modeling, 2024, arXiv preprint [arXiv:2401.16380](https://arxiv.org/abs/2401.16380).
- [29] MetaAI, Introducing meta llama 3: The most capable openly available llm to date, 2024, URL: <https://ai.meta.com/blog/meta-llama-3/>.
- [30] R. Wang, W. Zhou, M. Sachan, Let's synthesize step by step: Iterative dataset synthesis with large language models by extrapolating errors from small models, 2023, arXiv preprint [arXiv:2310.13671](https://arxiv.org/abs/2310.13671).
- [31] J. Li, K.-Y. Huang, J. Jin, J. Shi, A survey on statistical methods for health care fraud detection, *Heal. Care Manag. Sci.* 11 (2008) 275–287.
- [32] J.M. Johnson, T.M. Khoshgoftaar, Medicare fraud detection using neural networks, *J. Big Data* 6 (1) (2019) 63.
- [33] J. Hancock, T.M. Khoshgoftaar, Medicare fraud detection using catboost, in: 2020 IEEE 21st International Conference on Information Reuse and Integration for Data Science, IRI, IEEE, 2020, pp. 97–103.
- [34] C. Zhang, X. Xiao, C. Wu, Medical fraud and abuse detection system based on machine learning, *Int. J. Environ. Res. Public Heal.* 17 (19) (2020) 7265.
- [35] I. Matloob, S.A. Khan, H.U. Rahman, Sequence mining and prediction-based healthcare fraud detection methodology, *IEEE Access* 8 (2020) 143256–143273.
- [36] A. Bhaskar, S. Pande, R. Malik, A. Khamparia, et al., An intelligent unsupervised technique for fraud detection in health care systems, *Intell. Decis. Technol.* 15 (1) (2021) 127–139.
- [37] J.M. Johnson, T.M. Khoshgoftaar, Medical provider embeddings for healthcare fraud detection, *SN Comput. Sci.* 2 (4) (2021) 276.
- [38] K. Kapadiya, U. Patel, R. Gupta, M.D. Alshehri, S. Tanwar, G. Sharma, P.N. Bokoro, Blockchain and AI-empowered healthcare insurance fraud detection: an analysis, architecture, and future prospects, *IEEE Access* 10 (2022) 79606–79627.
- [39] G. Kalra, Y.K. Rajoria, R. Boadh, P. Rajendra, P. Pandey, N. Khatak, A. Kumar, Study of fuzzy expert systems towards prediction and detection of fraud case in health care insurance, *Mater. Today: Proc.* 56 (2022) 477–480.
- [40] I. Matloob, S.A. Khan, R. Rukaiya, M.A.K. Khattak, A. Munir, A sequence mining-based novel architecture for detecting fraudulent transactions in healthcare systems, *IEEE Access* 10 (2022) 48447–48463.
- [41] Y. Yoo, J. Shin, S. Kyeong, Medicare fraud detection using graph analysis: a comparative study of machine learning and graph neural networks, *IEEE Access* (2023).
- [42] J.M. Johnson, T.M. Khoshgoftaar, Data-centric ai for healthcare fraud detection, *SN Comput. Sci.* 4 (4) (2023) 389.
- [43] J. Lu, K. Lin, R. Chen, M. Lin, X. Chen, P. Lu, Health insurance fraud detection by using an attributed heterogeneous information network with a hierarchical attention mechanism, *BMC Med. Inform. Decis. Mak.* 23 (1) (2023) 62.
- [44] H. Shi, M.A. Tayebi, J. Pei, J. Cao, Cost-sensitive learning for medical insurance fraud detection with temporal information, *IEEE Trans. Knowl. Data Eng.* 35 (10) (2023) 10451–10463.
- [45] Y. Mao, Y. Li, B. Xu, J. Han, XGAN: A medical insurance fraud detector based on GAN with XGBoost, *J. Inf. Hiding Multim. Signal Process.* 15 (1) (2024) 36–52.
- [46] H. He, W. Graco, X. Yao, Application of genetic algorithm and k-nearest neighbour method in medical fraud detection, in: *Simulated Evolution and Learning: Second Asia-Pacific Conference on Simulated Evolution and Learning, SEAL'98 Canberra, Australia, November 24–27, 1998 Selected Papers 2*, Springer, 1999, pp. 74–81.
- [47] S.K. Majhi, S. Bhattacharya, R. Pradhan, S. Biswal, Fuzzy clustering using salp swarm algorithm for automobile insurance fraud detection, *J. Intell. Fuzzy Systems* 36 (3) (2019) 2333–2344.
- [48] C. Yan, M. Li, W. Liu, M. Qi, Improved adaptive genetic algorithm for the vehicle insurance fraud identification model based on a BP neural network, *Theoret. Comput. Sci.* 817 (2020) 12–23.
- [49] K. Parnian, F. Sorouri, A.N. Souza, A. Molazadeh, S. Mahdavi, Fraud detection in health insurance using a combination of feature subset selection based on squirrel optimization algorithm and nearest neighbors algorithm methods, *Futur. Gener. Distrib. Syst. J.* 3 (2) (2021) 1–11.
- [50] S.K. Majhi, Fuzzy clustering algorithm based on modified whale optimization algorithm for automobile insurance fraud detection, *Evol. Intell.* 14 (1) (2021) 35–46.
- [51] R. Lingeswari, S. Brindha, Efficient loss updated XGBoost with deep emended genetic algorithm for detecting online fraudulent transactions, *Multimedia Tools Appl.* (2024) 1–24.
- [52] V.N. Vapnik, An overview of statistical learning theory, *IEEE Trans. Neural Netw.* 10 (5) (1999) 988–999.
- [53] V. Vapnik, *The Nature of Statistical Learning Theory*, Springer science & business media, 2013.
- [54] G. James, D. Witten, T. Hastie, R. Tibshirani, et al., *An Introduction to Statistical Learning*, vol. 112, Springer, 2013.
- [55] S. Mirjalili, A. Lewis, The whale optimization algorithm, *Adv. Eng. Softw.* 95 (2016) 51–67.
- [56] Insurance Data, Hospital insurance claim fraud, 2023, URL: <https://www.kaggle.com/datasets/nikitkashyap/hospital-insurance-claim-fraud>.
- [57] Insurance Claim Data, Insurance claim - fraud detection with dataiku DSS, 2021, URL: <https://github.com/sulaihasubi/insurance-claims-fraud-detection/tree/main>.
- [58] S. Mirjalili, S. Mirjalili, Genetic algorithm, *Evol. Algorithms Neural Networks: Theory Appl.* (2019) 43–55.
- [59] S. Mirjalili, S.M. Mirjalili, A. Hatamlou, Multi-verse optimizer: a nature-inspired algorithm for global optimization, *Neural Comput. Appl.* 27 (2016) 495–513.
- [60] F. Marini, B. Walczak, Particle swarm optimization (PSO). a tutorial, *Chemometr. Intell. Lab. Syst.* 149 (2015) 153–165.
- [61] S. Mirjalili, A.H. Gandomi, S.Z. Mirjalili, S. Saremi, H. Faris, S.M. Mirjalili, Salp swarm algorithm: A bio-inspired optimizer for engineering design problems, *Adv. Eng. Softw.* 114 (2017) 163–191.
- [62] A.A. Heidari, S. Mirjalili, H. Faris, I. Aljarah, M. Mafarja, H. Chen, Harris hawks optimization: Algorithm and applications, *Future Gener. Comput. Syst.* 97 (2019) 849–872.